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Applications of Linear Theory

1. INTRODUCTION

The practical implications of linear filtering theory were soon recognized by the engineering community. Whereas classical least squares methods involve simultaneous processing of batches of observations with attendant data storage requirements, the filter operates on the data sequentially, requiring no data storage. The filter generates new state estimates as new observations become available, thus opening the possibility of real-time estimation. As a by-product, the filter generates the estimation error covariance matrix, which measures the uncertainty in the estimate.

The importance of having realistic estimation error variances cannot be overemphasized. Of what value is an estimate if we do not really know how good it is? Knowledge of the error variance is especially important if some decision is to be made on the basis of the estimate. Suppose we are journeying to the moon and we detect an error in our target destination. Should we correct our trajectory now or wait for more observations to be taken? How much should we correct? Should we correct now and again later? Answers to such questions as these depend not only on our estimate of the error, but also on the certainty of the error estimate.

Other important advantages accrue from the filter theory. It is possible,

for example, to perform a complete error analysis without actually simulating the filter. This is accomplished by solving the covariance matrix equations that are independent of the observations. As we shall subsequently see (Sections 3 and 7), the filter structure offers advantages over least squares in applications to nonlinear problems. Thus, although the linear filter is completely equivalent to least squares when the latter is properly interpreted (see Example 7.2), it offers numerous advantages in applications.

In the following paragraphs, we briefly sketch the applications of linear filter theory which have been made to aerospace engineering systems. Our sketch is no doubt incomplete, and the author apologizes in advance for any errors of omission. Although linear filter theory has been applied in other fields, the author is most familiar with aerospace applications. It is probably fair to say that most of the applications have been made by the aerospace community, and that this community has contributed significantly to the nontrivial task of reducing the theory to practice.

Perhaps the first practical application of linear filter theory was made by Smith, Schmidt, and their colleagues (see [22, 30, 35, 36]) to celestial navigation. Simultaneously, a similar application was made by Battin [4]. Pines *et al.* [23] applied the theory to orbit determination where, hitherto, least squares techniques had been in use. Controlled simulations indicated that the filter theory can be successfully applied to these problems. Such simulations are also reported by Smith [37]. We note that orbit determination problems are nonlinear, and standard linearization techniques (Section 3) were used in these applications.

Several problem areas were encountered by these investigators. One was the requirement of high precision in the filter computations. (See Section 13 for a technique that reduces numerical precision requirements.) The other, more fundamental problem encountered was that of modeling errors that can lead to divergence of the filter. Several investigators, Gunckel [14] and Pines *et al.* [24] among them, found that the filter did not perform as well under actual conditions (real observations) as it did under controlled conditions (simulated observations). This problem was already analyzed in Chapter 7, and will occupy much of our effort in the present chapter.

Following these early pioneering investigations, a number of authors report interesting applications of linear filter theory. These include the work of Rauch [28], Farrell [11], and Fitzgerald [12] in orbit determination and navigation, and the work of Wagner [39] in reentry trajectory estimation. By now the linear filter theory is relatively well known in the aerospace community, being the subject of survey papers and articles

(e.g., Schmidt [31] and Sorenson [38]). The Kalman–Bucy filter is now being used for ascent guidance at Cape Kennedy, and for various phases of the Apollo lunar mission [5]. It is also used in various places for error analyses.

The reduction of theory to practice is generally a long and difficult process. Such is the situation in the case of linear filter theory. Yet reports of applications often deal only with the successful final results and are thus, in a sense, very misleading, especially to the uninitiated. Schmidt *et al.* [34] recently reported the application of the theory to the C-5 aircraft navigation problem. This study goes into considerable detail in reporting the many problem areas encountered along the way in this application, and is recommended reading for the engineer interested in applications.

The Goddard study[1] of applications in orbit determination [23, 24] still continues. A recent status report on this work is given by Wolf [41]. After six years of effort (the author has been associated with this work for the past three years), the fundamental modeling problem still remains. Although several successful applications have been made, they have by no means been routine. Each new satellite presents new versions of the same problem. A dynamical model adequate for one satellite is inadequate for another one. This problem, of course, exists in all applications, but is particularly acute in tracking a satellite for many revolutions (orbits). These problems, and several solutions, are discussed in Sections 8–12.

This chapter deals with applications of linear filter theory to orbit determination and trajectory estimation problems. Since these problems are generally discrete (or continuous-discrete, which is essentially the same thing), we focus our attention here on the discrete filter. We review the discrete filter in Section 2 and outline its application to nonlinear problems in Section 3. The filter is then extended in Section 4 to systems with uncertain parameters which may, but need not, be estimated. In order to give the reader a "feel" for the filter, we treat a simple academic example in Section 5, giving results of numerical computations. The next two sections (Sections 6 and 7) present some applications of the linear filter in orbit determination and reentry trajectory estimation. The intent here is to show how the filter can perform in these systems under controlled conditions (simulated data). The next five sections deal with the important remaining problem area in filter applications, namely, the problem of filter divergence due to modeling errors. The

[1] This study is being conducted by the Special Projects Branch of the NASA Goddard Space Flight Center, under the direction of Mr. R. K. Squires.

problem is introduced and exhibited in Section 8. Sections 9 and 10 summarize various techniques that have been found useful in preventing divergence. In Sections 11 and 12, we show the applications of more sophisticated and promising techniques to the divergence problem. Finally, Section 13 briefly describes various miscellaneous developments in linear filter theory which are not treated in detail in this book.

2. REVIEW OF THE DISCRETE FILTER

Let us outline the computations involved in simulating the discrete linear filter. Our dynamical system model is (7.27, 7.2)

$$\begin{aligned} x_{k+1} &= \Phi(k+1, k)x_k + \Gamma(k)w_{k+1}, \\ y_k &= M(k)x_k + v_k, \end{aligned} \tag{8.1}$$

with

$$\begin{aligned} &\mathscr{E}\{w_k\} = 0, \qquad \mathscr{E}\{w_k w_l^{\mathrm{T}}\} = Q(k)\,\delta_{kl}, \\ &\mathscr{E}\{v_k\} = 0, \qquad \mathscr{E}\{v_k v_l^{\mathrm{T}}\} = R(k)\,\delta_{kl}, \qquad R(k) > 0. \end{aligned}$$

Refer to Section 7.3 for a more complete description of (8.1). Recall that $Y_k = \{y_1, \ldots, y_k\}$.

TABLE 8.1

FILTER VARIABLES

Variable	Definition	Dimension
$\hat{x}(k \mid k)$	State estimate at t_k given Y_k	$n \times 1$
$P(k \mid k)$	Covariance matrix of the error in $\hat{x}(k \mid k)$	$n \times n$
$\Phi(k+1 \mid k)$	State transition matrix (from t_k to t_{k+1})	$n \times n$
$\Gamma(k)$	System noise coefficient matrix	$n \times r$
$Q(k+1)$	System noise covariance matrix	$r \times r$
$\hat{x}(k+1 \mid k)$	State estimate at t_{k+1} given Y_k	$n \times 1$
$P(k+1 \mid k)$	Covariance matrix of the error in $\hat{x}(k+1 \mid k)$	$n \times n$
$M(k+1)$	Measurement matrix	$m \times n$
$R(k+1)$	Measurement noise covariance matrix	$m \times m$
$K(k+1)$	Filter (Kalman) gain matrix at t_{k+1}	$n \times m$
y_{k+1}	Measurement (observation) at t_{k+1}	$m \times 1$

With the variables of the filter described in Table 8.1, the filter computations can proceed as follows (see Theorem 7.2):

(1) Store the filter state $[\hat{x}(k \mid k), P(k \mid k)]$;

(2) Compute the predicted state

$$\hat{x}(k+1 \mid k) = \Phi(k+1, k)\, \hat{x}(k \mid k); \tag{8.2}$$

(3) Compute the predicted error covariance matrix

$$P(k+1 \mid k) = \Phi(k+1, k)\, P(k \mid k)\, \Phi^{\mathrm{T}}(k+1, k) + \Gamma(k) Q(k+1)\, \Gamma^{\mathrm{T}}(k); \tag{8.3}$$

(4) Compute the filter gain matrix

$$\begin{aligned} K(k+1) = {} & P(k+1 \mid k)\, M^{\mathrm{T}}(k+1) \\ & \times [M(k+1)\, P(k+1 \mid k)\, M^{\mathrm{T}}(k+1) + R(k+1)]^{-1}; \end{aligned} \tag{8.4}$$

(5) Process the observation y_{k+1}

$$\hat{x}(k+1 \mid k+1) = \hat{x}(k+1 \mid k) + K(k+1)[y_{k+1} - M(k+1)\, \hat{x}(k+1 \mid k)]; \tag{8.5}$$

(6) Compute the new error covariance matrix by

$$P(k+1 \mid k+1) = [I - K(k+1)\, M(k+1)]\, P(k+1 \mid k), \tag{8.6}$$

or by

$$\begin{aligned} P(k+1 \mid k+1) = {} & [I - K(k+1)\, M(k+1)] \\ & \times P(k+1 \mid k)[I - K(k+1)\, M(k+1)]^{\mathrm{T}} \\ & + K(k+1)\, R(k+1)\, K^{\mathrm{T}}(k+1); \end{aligned} \tag{8.7}$$

(7) Set $k = k + 1$, and return to step (1).

We have listed two different, although equivalent, equations for updating the error covariance matrix at an observation, namely, (8.6) and (8.7). As Aoki [2] points out, Eq. (8.7) is preferable for two reasons. First, the right-hand side of (8.7) is the sum of two symmetric, positive definite matrices, and, when these are added, the sum will be positive definite. On the other hand, (8.6) is, at best, the difference of two positive definite matrices. As a consequence, (8.7) is better conditioned for numerical computations, and will tend to retain more faithfully the positive definiteness and symmetry of $P(k+1 \mid k+1)$. As a second consideration, suppose a small error δK is made in the computation of the filter gain K. Then, to first order, we have from (8.6)

$$\delta P(k+1 \mid k+1) = -\delta K(k+1)\, M(k+1)\, P(k+1 \mid k),$$

whereas from (8.7), to first order,

$$\delta P(k+1 \mid k+1) = 0.$$

Thus, to first order, (8.7) is insensitive to errors in the filter gain and is to be preferred in numerical computations.

We mentioned, in passing, in Section 7.2 that

$$r(k+1 \mid k) \triangleq y_{k+1} - \mathscr{E}\{y_{k+1} \mid Y_k\} = y_{k+1} - M(k+1)\,\hat{x}(k+1 \mid k) \quad (8.8)$$

is called the *predicted residual* (more precisely, the predicted measurement residual error). The correction to the state estimate [see (8.5)] is proportional to the predicted residual, the constant of proportionality being the filter gain. It is easy to compute

$$\mathscr{E}\{r(k+1 \mid k)\} = 0, \quad (8.9)$$

and

$$\begin{aligned} Y(k+1 \mid k) &\triangleq \mathscr{E}\{r(k+1 \mid k)\, r^{\mathrm{T}}(k+1 \mid k)\} \\ &= M(k+1)\, P(k+1 \mid k)\, M^{\mathrm{T}}(k+1) + R(k+1). \end{aligned} \quad (8.10)$$

We see that the filter gain is proportional to the inverse of Y. Predicted residuals provide us with a useful tool for judging the performance of the filter in actual practice. These residuals are available to us, as are their statistics (8.9), (8.10). By checking whether residuals indeed possess their (theoretical) statistical properties, we are able to assess the performance of the filter. This is exploited in the adaptive filter, which we develop in Section 11.

The filter computations, as outlined in the foregoing, involve the inversion of the $m \times m$ residual covariance matrix Y. Recall that m is the dimension of the measurement vector. We have already indicated in Section 7.2 that this inversion can be avoided, under certain conditions, by processing the components of the measurement vector one at a time. More generally, suppose the measurement vector is partitioned as

$$y_{k+1} = \begin{bmatrix} \overset{(m_1 \times 1)}{y^1_{k+1}} \\ \vdots \\ \overset{(m_l \times 1)}{y^l_{k+1}} \end{bmatrix} = \begin{bmatrix} \overset{(m_1 \times n)}{M^1(k+1)}\, x_{k+1} + \overset{(m_1 \times 1)}{v^1_{k+1}} \\ \vdots \\ \overset{(m_l \times n)}{M^l(k+1)}\, x_{k+1} + \overset{(m_l \times 1)}{v^l_{k+1}} \end{bmatrix},$$

and that

$$\mathscr{E}\left\{ \begin{bmatrix} v^1_{k+1} \\ \vdots \\ v^l_{k+1} \end{bmatrix} [v^{1\mathrm{T}}_{k+1}, \ldots, v^{l\mathrm{T}}_{k+1}] \right\} = \begin{bmatrix} \overset{(m_1 \times m_1)}{R^1(k+1)} & 0 & \cdots & \cdots & 0 \\ & \overset{(m_2 \times m_2)}{} & & & \\ 0 & R^2(k+1) & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ & & & & \overset{(m_l \times m_l)}{} \\ 0 & \cdots & 0 & & R^l(k+1) \end{bmatrix};$$

that is, $\{v_{k+1}^1, \ldots, v_{k+1}^l\}$ are uncorrelated. Then, instead of processing the measurement vector y_{k+1} at once, we may equivalently process $y_{k+1}^1, \ldots, y_{k+1}^l$ one at a time. This is accomplished by iterating the computational procedure [steps (1)–(7)] l times with $\Phi \equiv I$, $\Gamma \equiv Q \equiv 0$. On the ith iteration, we process y_{k+1}^i, where $M(k+1)$ is replaced by $M^i(k+1)$, $R(k+1)$ by $R^i(k+1)$, and y_{k+1} by y_{k+1}^i. Thus on the ith iteration we invert

$$Y^i(k+1 \mid k) = M^i(k+1)\, P(k+1 \mid k)\, M^{i^{\mathrm{T}}}(k+1) + R^i(k+1),$$

which is an $m_i \times m_i$ matrix. [During the iterations, we clearly do not increase the index k in step (7).] In this procedure, the inversion of one $m \times m$ matrix is replaced by the inversion of l matrices of dimensions $m_1 \times m_1$, $m_2 \times m_2, \ldots, m_l \times m_l$. In case $l = m$ (the components of v_{k+1} are uncorrelated), we need only invert m scalars.

In concluding this section, we note that the equivalent "inverse" form of the filter, given in (7.11), can be used. This form, however, involves the inversion of $n \times n$ covariance matrices and is generally computationally inferior to the Kalman filter. If $m > n$, however, and the measurement noise components are correlated, then the inverse form can offer computational advantages.

3. EXTENSION TO NONLINEAR PROBLEMS

As was noted in the Introduction, orbit determination (and most other) problems are nonlinear and must be linearized (approximated) before the linear filter theory can be applied. We proceed to this task now. We shall deal with continuous nonlinear systems and discrete nonlinear observations, as this is the structure of the orbit determination problem (see Appendix 8A).

Suppose our nonlinear system is described by the stochastic n-vector differential equation

$$dx_t/dt = f(x_t, t) + G(t)w_t, \qquad t \geqslant t_0, \qquad x_{t_0} \sim N(\hat{x}_{t_0}, P_{t_0}), \tag{8.11}$$

where $\{w_t\}$ is a zero-mean, white Gaussian noise process with

$$\mathscr{E}\{w_t w_\tau^{\mathrm{T}}\} = Q(t)\,\delta(t-\tau).$$

Now suppose that we generate a *reference* (or *nominal*) deterministic trajectory $\bar{x}(t)$, with given $\bar{x}(t_0)$, satisfying

$$d\bar{x}(t)/dt = f(\bar{x}(t), t), \qquad t \geqslant t_0. \tag{8.12}$$

Define

$$\delta x_t \triangleq x_t - \bar{x}(t), \tag{8.13}$$

the *deviation* (or *perturbation*, or *variation*) from the reference trajectory. Then we see that $\{\delta x_t\}$ is a stochastic process satisfying the differential equation

$$d(\delta x_t)/dt = f(x_t, t) - f(\bar{x}(t), t) + G(t)w_t, \qquad \delta x_{t_0} \sim N(\hat{x}_{t_0} - \bar{x}(t_0), P_{t_0}). \tag{8.14}$$

If the deviations from the reference trajectory are "small" (say in the mean square sense), then a Taylor series expansion gives

$$f(x_t, t) - f(\bar{x}(t), t) \cong F[t; \bar{x}(t_0)]\, \delta x_t, \tag{8.15}$$

where

$$F[t; \bar{x}(t_0)] \triangleq \left[\frac{\partial f_i(\bar{x}(t), t)}{\partial x_j}\right] \tag{8.16}$$

is the matrix of partial derivatives evaluated along the reference trajectory. Thus we obtain the *approximate linear* equation (called the *perturbation* or *variational* equation)

$$d(\delta x_t)/dt = F[t; \bar{x}(t_0)]\, \delta x_t + G(t)w_t, \qquad \delta x_{t_0} \sim N(\hat{x}_{t_0} - \bar{x}(t_0), P_{t_0}). \tag{8.17}$$

We have emphasized the dependence of δx_t on the choice of the nominal trajectory by including $\bar{x}(t_0)$ in the argument of F. It should be understood, however, that F is a function of time only, being evaluated at the values $\bar{x}(t)$.

Now, as we did in Eqs. (7.19)–(7.26) of Section 7.2, we can discretize (8.17) as

$$\delta x_{t_{k+1}} = \Phi(t_{k+1}, t_k; \bar{x}(t_k))\, \delta x_{t_k} + w_{t_{k+1}}, \qquad \delta x_{t_0} \sim N(\hat{x}_{t_0} - \bar{x}(t_0), P_{t_0}), \tag{8.18}$$

where Φ is the state transition matrix [defined in (7.20)], and $\{w_{t_k}\}$ is a zero-mean, white Gaussian sequence, $w_{t_{k+1}} \sim N[0, Q(k+1)]$, where

$$Q(k+1) = \int_{t_k}^{t_{k+1}} \Phi(t_{k+1}, \tau)\, G(\tau)\, Q(\tau)\, G^{\mathrm{T}}(\tau)\, \Phi^{\mathrm{T}}(t_{k+1}, \tau)\, d\tau.$$

We, of course, now have the problem of computing the state transition matrix. We do not deal with that problem in this book. A good account of exact and approximate methods of computing the state transition matrix in orbit determination problems may be found in Lee *et al.* [21]. Our reason for discretizing the dynamics is that we prefer to compute the state transition matrix, rather than solving the variance equation

(7.99). It is noted in passing that the dynamics of satellite motion are usually considered to be deterministic ($w_{t_k} \equiv 0$). The noise input is, however, a useful, although artificial, method of accounting for computational errors, neglected nonlinearities [in (8.15)], and model errors. We shall have much more to say on this subject in Sections 8, 9, and 11.

We now turn to the measurement equation, which is assumed of the form

$$y_{t_k} = h(x_{t_k}, t_k) + v_k. \tag{8.19}$$

Defining the *nominal measurement* as

$$\bar{y}(t_k) \triangleq h(\bar{x}(t_k), t_k), \tag{8.20}$$

and

$$\delta y_{t_k} \triangleq y_{t_k} - \bar{y}(t_k), \tag{8.21}$$

and performing a similar linearization, we get the *linearized measurement equation*

$$\delta y_{t_k} = M[t_k ; \bar{x}(t_k)]\, \delta x_{t_k} + v_k, \tag{8.22}$$

where

$$M[t_k ; \bar{x}(t_k)] \triangleq \left[\frac{\partial h_i(\bar{x}(t_k), t_k)}{\partial x_j}\right]. \tag{8.23}$$

Let us summarize. We started with the *nonlinear* system

$$\begin{aligned} dx_t/dt &= f(x_t, t) + G(t)w_t, \qquad t \geqslant t_0, \qquad x_{t_0} \sim N(\hat{x}_{t_0}, P_{t_0}), \\ y_{t_k} &= h(x_{t_k}, t_k) + v_k, \end{aligned} \tag{8.24}$$

linearized about $\bar{x}(t)$, the solution of (8.12), and obtained the *linearized* discrete system

$$\begin{aligned} \delta x_{t_{k+1}} &= \Phi[t_{k+1}, t_k ; \bar{x}(t_k)]\, \delta x_{t_k} + w_{t_{k+1}}, \qquad \delta x_{t_0} \sim N(\hat{x}_{t_0} - \bar{x}(t_0), P_{t_0}), \\ \delta y_{t_k} &= M[t_k ; \bar{x}(t_k)]\, \delta x_{t_k} + v_k. \end{aligned} \tag{8.25}$$

Comparing (8.25) with (8.1), we see that the linear filter is directly applicable to the linearized system. Instead of "state" and "measurement," we now speak of the state deviation and measurement deviation. Given a reference trajectory and measurements y_{t_k}, we can compute δy_{t_k} [via (8.21)] and process the measurement deviations through the linear filter to estimate the state deviations. Then, from (8.13),

$$\hat{x}(t_k \mid t_k) = \bar{x}(t_k) + \widehat{\delta x}(t_k \mid t_k) \qquad (\widehat{\delta x} = \delta \hat{x}),$$

and

$$\tilde{x}(t_k \mid t_k) \triangleq x_{t_k} - \hat{x}(t_k \mid t_k) = \delta x_{t_k} - \delta\hat{x}(t_k \mid t_k) = \delta\tilde{x}(t_k \mid t_k),$$

so that

$$P(k \mid k) = \text{cov}\{\tilde{x}(t_k \mid t_k)\} = \text{cov}\{\delta\tilde{x}(t_k \mid t_k)\}.$$

The remaining problems are the choice of the reference trajectory and the question of the validity of the linearized equations. The latter question will occupy us in Chapter 9. Here we mention only that we expect the linearized dynamics to represent a good approximation if the system noise covariance (Q) and the initial uncertainty (P_{t_0}) are sufficiently small. For then, with the choice $\bar{x}(t_0) = \hat{x}_{t_0}$, the state deviations will remain small with a high probability, or in the mean square sense [see (3.132) and (4.164)]. Of course, what we have said is really incomplete, since it is clear from (8.25) that the size of the state deviations is also dependent upon the stability properties of the linearized system. Discussion of stochastic stability is beyond the scope of this book, and the interested reader is referred to Kushner [20]. Foreseeing the results of Chapter 9, we expect the linearized measurement equation to represent a good approximation if the measurement noise covariance (R) is sufficiently large. For then the neglected nonlinearities will be small compared to the noise, provided, of course, that the state deviations are small.

The obvious choice of the reference trajectory is made with $\bar{x}(t_0) = \hat{x}_{t_0}$, the prior estimate of the state. Then

$$\delta x_{t_0} \sim N(0, P_{t_0}),$$

and, as is evident from (8.25),

$$\mathscr{E}\{\delta x_{t_k}\} = 0, \qquad \text{all} \quad t_k .$$

But then, as we process the observations, obviously

$$\delta\hat{x}(t_k \mid t_k) = \mathscr{E}\{\delta x_{t_k} \mid Y_k\} \neq 0, \qquad k > 0,$$

in general, and our initially wise choice of a reference trajectory may turn out to be a poor one. This will be especially true if P_{t_0} is large (so that $\hat{x}_{t_0}$ is not a good estimate), and/or if the system noise (Q) is large. In that case, the estimates of the state deviations can become large, violating our linearity assumptions.

Suppose that our dynamical system is unforced by noise[2] ($w_t \equiv 0$).

[2] The methods we are about to develop are probably good even in the presence of small system noise, but our intuitive arguments fail in that case.

Then we can imagine that there exists a "true" trajectory of the system, which we could learn if we had enough observations available (see Lemma 7.1 and the discussion following). Suppose that, having chosen $\bar{x}(t_0) = \hat{x}_{t_0}$, we filter the batch of observations

$$\{y_1, \cdots, y_k\}$$

and obtain $\delta\hat{x}(t_k \mid t_k)$. Then, predicting backward to t_0,

$$\delta\hat{x}(t_0 \mid t_k) = \Phi[t_0, t_k ; \bar{x}(t_0)]\, \delta\hat{x}(t_k \mid t_k).$$

If our batch of observations is sufficiently large, then we can expect

$$\bar{x}'(t_0) = \bar{x}(t_0) + \delta\hat{x}(t_0 \mid t_k) \tag{8.26}$$

to be closer to the "true" initial state (x_{t_0}) than $\bar{x}(t_0)$ is. Taking $\bar{x}'(t_0)$ now as the reference initial condition, we can reprocess our batch of observations, that is, linearize about $\bar{x}'(t)$ and run the filter for the linearized system over the same batch of observations.[3] This procedure can be repeated several times (iterated) until (8.26) produces no change in the reference initial condition. Presumably, we shall have then converged to the true trajectory, or as close to the true trajectory as the information in our batch of observations and neglected nonlinear effects allow.

We shall call the procedure just described *global iteration* of the Kalman filter. This type of iteration is commonly used in batch (least squares) estimation techniques. In fact, in view of the equivalence of least squares and the (recursive) filter shown in Example 7.2, the global iteration we have described is equivalent to iterated least squares. It should be noted that global iteration destroys the advantages of the filter as described in the introductory section. Furthermore, there is no guarantee that the iteration will converge. In fact, the iteration may diverge, requiring alternate guesses of the reference trajectory.

With the linear filter recursive structure, we can go a step further. We can relinearize about each new estimate as new estimates become available. This procedure goes as follows. At t_0, linearize about $\hat{x}_{t_0}$. Once y_1 is processed, relinearize about $\hat{x}(t_1 \mid t_1)$, and so on. The point of this is to use a better reference trajectory as soon as one is available. As a consequence of relinearization, large initial estimation errors are not allowed to propagate through time, and, therefore, our linearity assumptions are less likely to be violated. We note that this advantage is not available in least squares batch estimation techniques.

[3] When running the filter for the second time, we *do not* change the statistics of x_{t_0}. We only change the reference trajectory.

Now if we initially linearize about $\hat{x}_{t_0}$, then

$$\delta\hat{x}(t_0 \mid t_0) = 0,$$

and, in view of (8.2) (applied to the deviations),

$$\delta\hat{x}(t_1 \mid t_0) = 0.$$

Since we subsequently linearize about $\hat{x}(t_1 \mid t_1)$,

$$\delta\hat{x}(t_1 \mid t_1) = 0,$$

so that again

$$\delta\hat{x}(t_2 \mid t_1) = 0;$$

and, in general,

$$\delta\hat{x}(t \mid t_k) = 0, \qquad t_k \leqslant t \leqslant t_{k+1}, \qquad \text{all} \quad k. \tag{8.27}$$

As a result, between observations the best estimate of the state is the reference state, and, accordingly,

$$d\hat{x}(t \mid t_k)/dt = f(\hat{x}(t \mid t_k), t), \qquad t_k \leqslant t \leqslant t_{k+1}. \tag{8.28}$$

Prediction is accomplished using the *nonlinear* system differential equation (8.24) with $w_t \equiv 0$. Compare this with Example 7.12. We shall have more to say about this in Chapter 9.

Now write the equation for the correction to the estimate at an observation (8.5) for the linearized system

$$\delta\hat{x}(k+1 \mid k+1) = \delta\hat{x}(k+1 \mid k) + K(k+1)[\delta y_{k+1} - M(k+1)\,\delta\hat{x}(k+1 \mid k)].$$

Since

$$\delta\hat{x}(k+1 \mid k+1) = \hat{x}(k+1 \mid k+1) - \hat{x}(k+1 \mid k)$$

in view of relinearization, and using (8.27), this becomes

$$\hat{x}(k+1 \mid k+1) = \hat{x}(k+1 \mid k) + K(k+1)[y_{k+1} - h(\hat{x}(k+1 \mid k), t_{k+1})]. \tag{8.29}$$

We have just derived what is known as the *extended* (or *modified*) Kalman filter for the nonlinear system (8.24). This filter, the result of the relinearization procedure described, consists of the computations outlined in steps (1)–(7) of Section 2, but with (8.2) replaced by (8.28), and (8.5) replaced by (8.29). The extended Kalman filter is commonly used in orbit determination. We summarize it in the following theorem.

Theorem 8.1 (Extended Kalman Filter). *The extended Kalman filter for the nonlinear system* (8.24) *consists of prediction via*

$$\hat{x}(t_{k+1} \mid t_k) = \hat{x}(t_k \mid t_k) + \int_{t_k}^{t_{k+1}} f(\hat{x}(t \mid t_k), t)\, dt, \tag{8.30}$$

$$P(t_{k+1} \mid t_k) = \Phi[t_{k+1}, t_k; \hat{x}(t_k \mid t_k)]\, P(t_k \mid t_k)\, \Phi^{\mathrm{T}}[t_{k+1}, t_k; \hat{x}(t_k \mid t_k)] + Q(t_{k+1}); \tag{8.31}$$

and at an observation

$$\begin{aligned}\hat{x}(t_{k+1} \mid t_{k+1}) &= \hat{x}(t_{k+1} \mid t_k) + K[t_{k+1}; \hat{x}(t_{k+1} \mid t_k)] \\ &\quad \times [y_{t_{k+1}} - h(\hat{x}(t_{k+1} \mid t_k), t_{k+1})],\end{aligned} \tag{8.32}$$

$$\begin{aligned}P(t_{k+1} \mid t_{k+1}) &= [I - K\{t_{k+1}; \hat{x}(t_{k+1} \mid t_k)\}\, M\{t_{k+1}; \hat{x}(t_{k+1} \mid t_k)\}]\, P(t_{k+1} \mid t_k) \\ &\quad \times [I - K\{t_{k+1}; \hat{x}(t_{k+1} \mid t_k)\}\, M\{t_{k+1}; \hat{x}(t_{k+1} \mid t_k)\}]^{\mathrm{T}} \\ &\quad + K[t_{k+1}; \hat{x}(t_{k+1} \mid t_k)]\, R(k+1)\, K^{\mathrm{T}}[t_{k+1}; \hat{x}(t_{k+1} \mid t_k)].\end{aligned} \tag{8.33}$$

The Kalman gain is

$$\begin{aligned}K[t_{k+1}; \hat{x}(t_{k+1} \mid t_k)] &= P(t_{k+1} \mid t_k)\, M^{\mathrm{T}}[t_{k+1}; \hat{x}(t_{k+1} \mid t_k)] \\ &\quad \times [M\{t_{k+1}; \hat{x}(t_{k+1} \mid t_k)\}\, P(t_{k+1} \mid t_k) \\ &\quad \times M^{\mathrm{T}}\{t_{k+1}; \hat{x}(t_{k+1} \mid t_k)\} + R(k+1)]^{-1}.\end{aligned} \tag{8.34}$$

The matrices Φ and M are those of the linearized system (8.25). We have emphasized the fact that these matrices depend on $\hat{x}(t_k \mid t_k)$, or, equivalently, on $\hat{x}(t_{k+1} \mid t_k)$, by including these estimates as arguments. We shall often omit these arguments for convenience of notation.

We next describe some *local* iteration algorithms that are based largely on the extended Kalman filter. By local iteration, we mean iteration at a point t_{k+1}, or on an interval $[t_k, t_{k+1}]$. The purpose of the iterations is to improve the reference trajectory, and thus the estimate, in the presence of significant nonlinearities. Since the iteration is local, the recursive filter structure is retained; new estimates are computed as new observations become available.

Consider the estimator, Eq. (8.32), in the extended Kalman filter. We obtained (8.32) by evaluating (8.5) (for the linearized system),

$$\begin{aligned}\delta\hat{x}(t_{k+1} \mid t_{k+1}) &= \delta\hat{x}(t_{k+1} \mid t_k) + K[t_{k+1}; \bar{x}(t_{k+1})] \\ &\quad \times [\delta y_{k+1} - M\{t_{k+1}; \bar{x}(t_{k+1})\}\, \delta\hat{x}(t_{k+1} \mid t_k)],\end{aligned} \tag{8.35}$$

about $\bar{x}(t_{k+1}) = \hat{x}(t_{k+1} \mid t_k)$ as reference. Then, processing the observa-

tion y_{k+1} via (8.32), we get $\hat{x}(t_{k+1} \mid t_{k+1})$. Now $\hat{x}(t_{k+1} \mid t_{k+1})$ is a "better" estimate (presumably closer to the true trajectory) than $\hat{x}(t_{k+1} \mid t_k)$. Let us relinearize about $\hat{x}(t_{k+1} \mid t_{k+1})$ then, and recompute the estimate. In general, evaluate (8.35) about η_i, and call the output of (8.35) η_{i+1} to get the iterator

$$\eta_{i+1} = \hat{x}(t_{k+1} \mid t_k) + K(t_{k+1}\,;\eta_i)[y_{k+1} - h(\eta_i\,, t_{k+1}) - M(t_{k+1}\,;\eta_i) \times \{\hat{x}(t_{k+1} \mid t_k) - \eta_i\}], \tag{8.36}$$

$i = 1, 2,\ldots$; $\eta_1 = \hat{x}(t_{k+1} \mid t_k)$. Note that η_2 is just the result of the extended Kalman filter (8.32). Note also that the gain K is reevaluated on each iteration, as are the measurement function h and the matrix M. The iteration is terminated when there is no significant change in consecutive iterates. Then the last iterate, η_l say, is taken for the estimate $\hat{x}(t_{k+1} \mid t_{k+1})$.

This iterated extended Kalman filter is apparently due to J. V. Breakwell. It is presented and analyzed by Denham and Pines [7], who show that it can be useful in reducing the effect of measurement function (h) nonlinearity, thus improving the linear filter performance. We shall have more to say about this iterator in Chapter 9. For now, we summarize it in the following theorem.

Theorem 8.2 (Iterated Extended Kalman Filter). *The iterated extended Kalman filter consists of Theorem* 8.1 *with* (8.32) *replaced by the iterator*:

$$\eta_{i+1} = \hat{x}(t_{k+1} \mid t_k) + K(t_{k+1}\,;\eta_i)[y_{k+1} - h(\eta_i\,, t_{k+1}) - M(t_{k+1}\,;\eta_i) \times \{\hat{x}(t_{k+1} \mid t_k) - \eta_i\}], \qquad i = 1,\ldots, l, \tag{8.36}$$

$$\hat{x}(t_{k+1} \mid t_{k+1}) = \eta_l\,. \tag{8.37}$$

The iteration starts with $\eta_1 = \hat{x}(t_{k+1} \mid t_k)$, *and terminates when there is no significant difference between consecutive iterates. The covariance matrix in* (8.33) *is then computed based on* η_l, *that is, based on linearization about* η_l.

The preceding iterator is designed for measurement nonlinearities and does not improve the previous reference trajectory on $[t_k\,, t_{k+1})$. Thus, $\hat{x}(t_{k+1} \mid t_k)$ can be in error due to system nonlinearities acting on the interval $[t_k\,, t_{k+1})$. Of course, the reference trajectory on $[t_{k+1}\,, t_{k+2})$ will be improved. We can, in the following way, include the reference trajectory $[t_k\,, t_{k+1})$ in the iteration loop. We start at t_k with $\hat{x}(t_k \mid t_k)$ and $P(t_k \mid t_k)$. We linearize and predict to t_{k+1} and apply one iteration of (8.36). Based on that new estimate, we smooth back to t_k. The smoothing closes the loop, providing an improved reference for prediction to t_{k+1}. Let us derive the equations for this iterator.

We are at t_k with $\hat{x}(t_k \mid t_k)$ and $P(t_k \mid t_k)$. Linearize about ξ_i. Then, from (8.12),

$$\bar{x}(t_{k+1} ; \xi_i) = \xi_i + \int_{t_k}^{t_{k+1}} f(\bar{x}(\tau), \tau)\, d\tau, \tag{8.38}$$

and, from (8.2) for the linearized system,

$$\delta\hat{x}(t_{k+1} \mid t_k) = \Phi(t_{k+1}, t_k ; \xi_i)\, \delta\hat{x}(t_k \mid t_k),$$

or

$$\hat{x}(t_{k+1} \mid t_k) = \bar{x}(t_{k+1} ; \xi_i) + \Phi(t_{k+1}, t_k ; \xi_i)[\hat{x}(t_k \mid t_k) - \xi_i]. \tag{8.39}$$

Now apply iterator (8.36) to process the observation and get η_{i+1}. We now have the following choice. Linearize about η_{i+1} to smooth back to t_k [this involves solving (8.12) backward in time], or keep $\bar{x}(t, \xi_i)$ as reference for the smoothing. We make the latter choice, since we shall improve $\bar{x}(t, \xi_i)$ after the smoothing anyway. From Example 7.8, the linearized smoother is

$$\delta\hat{x}(t_k \mid t_{k+1}) = \delta\hat{x}(t_k \mid t_k) + S(t_k ; \xi_i)[\delta\hat{x}(t_{k+1} \mid t_{k+1}) - \delta\hat{x}(t_{k+1} \mid t_k)],$$

with

$$S(t_k ; \xi_i) = P(t_k \mid t_k)\, \Phi^{\mathrm{T}}(t_{k+1}, t_k ; \xi_i)\, P^{-1}(t_{k+1} \mid t_k); \tag{8.40}$$

or, using the smoothed estimate as ξ_{i+1},

$$\xi_{i+1} = \hat{x}(t_k \mid t_k) + S(t_k ; \xi_i)[\eta_{i+1} - \hat{x}(t_{k+1} \mid t_k)]. \tag{8.41}$$

[Note that $P(t_{k+1} \mid t_k)$ in (8.40) depends on ξ_i.] ξ_{i+1} is our improved reference for the prediction in (8.38). This iteration starts with $\xi_1 = \hat{x}(t_k \mid t_k)$ and $\eta_1 = \hat{x}(t_{k+1} \mid t_k)$. Thus η_2 is the output of the extended Kalman filter.

This last iterator was apparently first derived by Wishner *et al.* [40], but in an entirely different way. We shall call it the *iterated linear filter-smoother*. Actually, the iterators we have derived are closely related in theory; both have a similar probabilistic interpretation that we shall pursue in Chapter 9. We summarize as follows:

Theorem 8.3 (Iterated Linear Filter-Smoother). *Given* $\hat{x}(t_k \mid t_k)$ *and* $P(t_k \mid t_k)$, *the iterated linear filter-smoother consists of the iteration*

$$\eta_{i+1} = \hat{x}(t_{k+1} \mid t_k) + K(t_{k+1} ; \eta_i, \xi_i)[y_{k+1} - h(\eta_i, t_{k+1}) - M(t_{k+1} ; \eta_i) \times \{\hat{x}(t_{k+1} \mid t_k) - \eta_i\}], \tag{8.42}$$

$$\xi_{i+1} = \hat{x}(t_k \mid t_k) + S(t_k ; \xi_i)[\eta_{i+1} - \hat{x}(t_{k+1} \mid t_k)], \qquad i = 1, \ldots, l, \tag{8.43}$$

$$\eta_1 = \hat{x}(t_{k+1} \mid t_k), \qquad \xi_1 = \hat{x}(t_k \mid t_k),$$

where

$$\bar{x}(t_{k+1}\,;\,\xi_i) = \xi_i + \int_{t_k}^{t_{k+1}} f(\bar{x}(\tau), \tau)\, d\tau, \tag{8.44}$$

$$\hat{x}(t_{k+1} \mid t_k) = \bar{x}(t_{k+1}\,;\,\xi_i) + \Phi(t_{k+1}, t_k\,;\,\xi_i)(\hat{x}(t_k \mid t_k) - \xi_i), \tag{8.45}$$

$$\begin{aligned} K(t_{k+1}\,;\,\eta_i, \xi_i) = {} & P(t_{k+1} \mid t_k)\, M^{\mathrm{T}}(t_{k+1}\,;\,\eta_i)[M(t_{k+1}\,;\,\eta_i)\, P(t_{k+1} \mid t_k) \\ & \times M^{\mathrm{T}}(t_{k+1}\,;\,\eta_i) + R(k+1)]^{-1}, \end{aligned} \tag{8.46}$$

$$P(t_{k+1} \mid t_k) = \Phi(t_{k+1}, t_k\,;\,\xi_i)\, P(t_k \mid t_k)\, \Phi^{\mathrm{T}}(t_{k+1}, t_k\,;\,\xi_i) + Q(t_{k+1}), \tag{8.47}$$

$$S(t_k\,;\,\xi_i) = P(t_k \mid t_k)\, \Phi^{\mathrm{T}}(t_{k+1}, t_k\,;\,\xi_i)\, P^{-1}(t_{k+1} \mid t_k). \tag{8.48}$$

Then

$$\hat{x}(t_{k+1} \mid t_{k+1}) = \eta_l\,, \tag{8.49}$$

$$\begin{aligned} P(t_{k+1} \mid t_{k+1}) = {} & [I - K(t_{k+1}\,;\,\eta_l, \xi_l)\, M(t_{k+1}\,;\,\eta_l)]\, P(t_{k+1} \mid t_k\,;\,\xi_l) \\ & \times [I - K(t_{k+1}\,;\,\eta_l, \xi_l)\, M(t_{k+1}\,;\,\eta_l)]^{\mathrm{T}} \\ & + K(t_{k+1}\,;\,\eta_l \quad \xi_l)\, R(k+1)\, K^{\mathrm{T}}(t_{k+1}\,;\,\eta_l, \xi_l). \end{aligned} \tag{8.50}$$

We note in closing this section that the extended Kalman filter may also be *iterated globally*. Having processed $\{y_1, \ldots, y_k\}$, starting with $\hat{x}_{t_0}$ and P_{t_0}, we have $\hat{x}(t_k \mid t_k)$ and $P(t_k \mid t_k)$. Then, assuming $w_t \equiv 0$, we solve (8.12) backward in time with $\hat{x}(t_k \mid t_k)$ as initial condition. Thus we get the smoothed estimate $\hat{x}(t_0 \mid t_k)$. We can then reprocess the observations, starting with $\hat{x}(t_0 \mid t_k)$, P_{t_0}. Note that in doing this we are changing the initial statistic $\mathscr{E}\{x_{t_0}\}$, which is, strictly speaking, part of the problem statement. This is, however, often justified by the fact that $\hat{x}_{t_0}$ is only an educated guess anyway.

4. UNCERTAIN PARAMETERS

Frequently, in applications, the dynamical system depends on certain parameters whose values are imprecisely known. Such parameters may be regarded as random variables with known *a priori* statistics. We model such a system containing uncertain parameters by

$$x_{k+1} = \Phi(k+1, k)x_k + \Psi(k+1, k)u + \Gamma(k)w_{k+1}\,, \tag{8.51}$$

$$y_k = M(k)x_k + N(k)p + v_k\,. \tag{8.52}$$

The parameters in the vector u will be referred to as *dynamical param-*

eters, whereas those in the vector p will be called *measurement parameters*. The *a priori* statistics of u and p will be taken as

$$\begin{aligned} \mathscr{E}\{u\} &= 0, & \mathscr{E}\{p\} &= 0, \\ \mathscr{E}\{uu^{\mathrm{T}}\} &= U_0, & \mathscr{E}\{pp^{\mathrm{T}}\} &= W_0. \end{aligned} \tag{8.53}$$

We assume that u, p, x_0, $\{w_k\}$, and $\{v_k\}$ are uncorrelated.

Now these (constant) parameters may be regarded as outputs of the dynamical systems

$$\begin{aligned} u_{k+1} &= u_k, & u_0 &= u, \\ p_{k+1} &= p_k, & p_0 &= p. \end{aligned} \tag{8.54}$$

Then, defining the *augmented* state vector,

$$X_k = \begin{bmatrix} x_k \\ u_k \\ p_k \end{bmatrix}, \tag{8.55}$$

we can combine the systems (8.51), (8.54) into the single *augmented* system

$$X_{k+1} = \begin{bmatrix} \Phi(k+1,k) & \Psi(k+1,k) & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix} X_k + \begin{bmatrix} \Gamma(k) \\ 0 \\ 0 \end{bmatrix} w_{k+1} \tag{8.56}$$

with observations

$$y_k = [M(k) \quad 0 \quad N(k)]X_k + v_k. \tag{8.57}$$

Now the linear filter is clearly directly applicable to this augmented system, and produces estimates

$$\hat{X}(k \mid k) = \begin{bmatrix} \hat{x}(k \mid k) \\ \hat{u}(k \mid k) \\ \hat{p}(k \mid k) \end{bmatrix}.$$

Thus, we can estimate these uncertain parameters, together with the original state x, in a straightforward manner.

Although straightforward in theory, the estimation of parameters in addition to the state increases the computational load and therefore may not be desirable. On the other hand, uncertain parameters cannot be ignored, since they can produce divergence of the filter (see Sections 7.8 and 8). An alternative, developed by Schmidt [31], is to take into account the effect of the uncertain parameters in degrading the estimate of the state, without estimating the parameters themselves. Such a filter can

be derived by first writing the standard linear filter for the augmented system (8.56), (8.57), and then throwing away the estimation equations for the parameters. That is, we shall insist that

$$\hat{u}(k \mid k) = \hat{p}(k \mid k) \equiv 0 = \mathscr{E}\{u\} = \mathscr{E}\{p\},$$

and

$$\mathscr{E}\{[u - \hat{u}(k \mid k)][u - \hat{u}(k \mid k)]^{\mathrm{T}}\} \equiv U_0,$$
$$\mathscr{E}\{[p - \hat{p}(k \mid k)][p - \hat{p}(k \mid k)]^{\mathrm{T}}\} \equiv W_0.$$

As a consequence, since u and p are initially uncorrelated, they will remain uncorrelated during the filtering process. The resulting filter will clearly be suboptimal, since the parameters are not being estimated. For if the parameters were estimated, then their uncertainty would decrease, permitting the uncertainty in the state estimate to decrease.

In view of the aforementioned *a posteriori* assumptions, the error covariance matrix $\mathscr{P}$ for the augmented system will be of the form

$$\mathscr{P} = \begin{bmatrix} P & C_u & C_p \\ C_u^{\mathrm{T}} & U_0 & 0 \\ C_p^{\mathrm{T}} & 0 & W_0 \end{bmatrix}, \tag{8.58}$$

where P is the usual error covariance matrix for the state x,

$$P(k \mid l) = \mathscr{E}\{[x_k - \hat{x}(k \mid l)][x_k - \hat{x}(k \mid l)]^{\mathrm{T}}\}, \tag{8.59}$$

and where

$$\begin{aligned} C_u(k \mid l) &= \mathscr{E}\{[x_k - \hat{x}(k \mid l)]u^{\mathrm{T}}\}, \\ C_p(k \mid l) &= \mathscr{E}\{[x_k - \hat{x}(k \mid l)]p^{\mathrm{T}}\}. \end{aligned} \tag{8.60}$$

Using (8.2) for the augmented system, we get that

$$\hat{x}(k+1 \mid k) = \Phi(k+1 \mid k)\,\hat{x}(k \mid k), \tag{8.61}$$

and $\hat{u}$, $\hat{p}$ are invariant in prediction. Then, from (8.3), for the augmented system

$$\mathscr{P}(k+1 \mid k) = \begin{bmatrix} \Phi(k+1,k) & \Psi(k+1,k) & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix} \begin{bmatrix} P(k \mid k) & C_u(k \mid k) & C_p(k \mid k) \\ C_u^{\mathrm{T}}(k \mid k) & U_0 & 0 \\ C_p^{\mathrm{T}}(k \mid k) & 0 & W_0 \end{bmatrix}$$
$$\times \begin{bmatrix} \Phi^{\mathrm{T}}(k+1,k) & 0 & 0 \\ \Psi^{\mathrm{T}}(k+1,k) & I & 0 \\ 0 & 0 & I \end{bmatrix} + \begin{bmatrix} \Gamma(k)Q(k+1)\Gamma^{\mathrm{T}}(k) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

which is

$$\begin{aligned} P(k+1 \mid k) = {} & \Phi(k+1,k)\, P(k \mid k)\, \Phi^{\mathrm{T}}(k+1,k) \\ & + \Phi(k+1,k)\, C_u(k \mid k)\, \Psi^{\mathrm{T}}(k+1,k) \\ & + \Psi(k+1,k)\, C_u^{\mathrm{T}}(k \mid k)\, \Phi^{\mathrm{T}}(k+1,k) \\ & + \Psi(k+1,k)\, U_0 \Psi^{\mathrm{T}}(k+1,k) \\ & + \Gamma(k)\, Q(k+1)\, \Gamma^{\mathrm{T}}(k), \end{aligned} \tag{8.62}$$

$$C_u(k+1 \mid k) = \Phi(k+1,k)\, C_u(k \mid k) + \Psi(k+1,k) U_0, \tag{8.63}$$

$$C_p(k+1 \mid k) = \Phi(k+1,k)\, C_p(k \mid k), \tag{8.64}$$

and U_0 and W_0 are invariant in prediction.

Let $\mathscr{M}(k)$ be the measurement matrix for the augmented system,

$$\mathscr{M}(k) = [M(k) \quad 0 \quad N(k)].$$

Then

$$\mathscr{P}(k+1 \mid k)\, \mathscr{M}^{\mathrm{T}}(k+1)$$

$$= \begin{bmatrix} P(k+1 \mid k)\, M^{\mathrm{T}}(k+1) + C_p(k+1 \mid k)\, N^{\mathrm{T}}(k+1) \\ C_u^{\mathrm{T}}(k+1 \mid k)\, M^{\mathrm{T}}(k+1) \\ C_p^{\mathrm{T}}(k+1 \mid k)\, M^{\mathrm{T}}(k+1) + W_0 N^{\mathrm{T}}(k+1) \end{bmatrix},$$

$$\begin{aligned} \mathscr{Y}(k+1 \mid k) \triangleq {} & \mathscr{M}(k+1)\, \mathscr{P}(k+1 \mid k)\, \mathscr{M}^{\mathrm{T}}(k+1) + R(k+1) \\ = {} & M(k+1)\, P(k+1 \mid k)\, M^{\mathrm{T}}(k+1) \\ & + M(k+1)\, C_p(k+1 \mid k)\, N^{\mathrm{T}}(k+1) \\ & + N(k+1)\, C_p^{\mathrm{T}}(k+1 \mid k)\, M^{\mathrm{T}}(k+1) \\ & + N(k+1)\, W_0 N^{\mathrm{T}}(k+1) + R(k+1). \end{aligned} \tag{8.65}$$

Now, from (8.5) for the augmented system,

$$\hat{x}(k+1 \mid k+1) = \hat{x}(k+1 \mid k) + \mathscr{K}(k+1)[y_{k+1} - M(k+1)\, \hat{x}(k+1 \mid k)], \tag{8.66}$$

where

$$\begin{aligned} \mathscr{K}(k+1) = {} & [P(k+1 \mid k)\, M^{\mathrm{T}}(k+1) \\ & + C_p(k+1 \mid k)\, N^{\mathrm{T}}(k+1)]\, \mathscr{Y}^{-1}(k+1 \mid k), \end{aligned} \tag{8.67}$$

and we throw away the estimation equations for u and p. From (8.6) for the augmented system,

$$P(k+1 \mid k+1) = P(k+1 \mid k) - \mathscr{K}(k+1)[M(k+1)\,P(k+1 \mid k) + N(k+1)\,C_p^{\mathrm{T}}(k+1 \mid k)], \tag{8.68}$$

$$C_u(k+1 \mid k+1) = C_u(k+1 \mid k) - \mathscr{K}(k+1)\,M(k+1)\,C_u(k+1 \mid k), \tag{8.69}$$

$$C_p(k+1 \mid k+1) = C_p(k+1 \mid k) - \mathscr{K}(k+1)[M(k+1)\,C_p(k+1 \mid k) + N(k+1)\,W_0], \tag{8.70}$$

and we keep U_0 and W_0 invariant at an observation.

We have derived what we shall call the *Schmidt–Kalman* filter with uncertain parameters. It is summarized in the following theorem.

Theorem 8.4 (Schmidt–Kalman Filter with Uncertain Parameters). *The system is* (8.51), (8.52), *where the uncertain parameters u and p have the statistics given in* (8.53). *The filter is for the state x only; the uncertain parameters are allowed only to degrade the estimate of the state, and are not being estimated themselves. This filter consists of equations for the state estimate* $\hat{x}(k \mid k)$, *the state estimation error covariance matrix P defined in* (8.59), *and the correlation matrices* C_u *and* C_p *defined in* (8.60). *Between observations, these quantities satisfy* (8.61), (8.62), (8.63), *and* (8.64), *respectively. At an observation, they satisfy* (8.66), (8.68), (8.69), *and* (8.70), *respectively.* $C_u(0 \mid 0) = C_p(0 \mid 0) = 0$.

The Schmidt–Kalman filter involves recursions for the correlation matrices C_u and C_p, in addition to the usual equations for $\hat{x}$ and P. Thus, the computational requirements are increased. However, they are considerably lower than the computational requirements for the augmented state filter. Thus, if improved estimates of the uncertain parameters are not required, the Schmidt–Kalman filter is an attractive alternative.

Example 8.1. Show, using the method of Example 7.4, that, under the stated *a posteriori* assumptions, the gain $\mathscr{K}$ in (8.67) is optimal. ▲

Example 8.2. Show directly from the system equation (8.51), and from (8.61) and (8.66), that the matrix P defined by (8.62) and (8.68) is indeed the estimation error covariance matrix, as defined in (8.59). That is, the covariance matrix computed in the filter is the covariance of filter estimation errors. ▲

The Schmidt–Kalman filter for uncertain parameters can be readily applied, by linearization, to the *nonlinear* system

$$\begin{aligned} dx_t/dt &= f(x_t, u, t) + G(t)w_t, \qquad t \geqslant t_0, \qquad x_{t_0} \sim N(\hat{x}_{t_0}, P_{t_0}), \\ y_{t_k} &= h(x_{t_k}, p, t_k) + v_k. \end{aligned} \tag{8.71}$$

Generate a reference trajectory with a given $\bar{x}(t_0)$, and, for a nominal value of u, call it $\bar{u}$, satisfying

$$d\bar{x}(t)/dt = f(\bar{x}(t), \bar{u}, t), \qquad t \geqslant t_0.$$

Then

$$d(\delta x_t)/dt = F(t; \bar{x}(t_0), \bar{u})\,\delta x_t + L(t; \bar{x}(t_0), \bar{u})\,\delta u + G(t)w_t, \tag{8.72}$$

where

$$F(t; \bar{x}(t_0), \bar{u}) \triangleq \left[\frac{\partial f_i(\bar{x}(t), \bar{u}, t)}{\partial x_j}\right], \qquad L(t; \bar{x}(t_0), \bar{u}) \triangleq \left[\frac{\partial f_i(\bar{x}(t), \bar{u}, t)}{\partial u_j}\right].$$

Equation (8.72) can be discretized

$$\begin{aligned} \delta x_{t_{k+1}} &= \Phi(t_{k+1}, t_k; \bar{x}(t_k), \bar{u})\,\delta x_{t_k} + \Psi(t_{k+1}, t_k; \bar{x}(t_k), \bar{u})\,\delta u + w_{k+1}, \\ \delta x_{t_0} &\sim N(\hat{x}_{t_0} - \bar{x}(t_0), P_{t_0}), \end{aligned} \tag{8.73}$$

where Φ is the usual state transition matrix and $\psi(t, \tau)$ satisfies

$$d\Psi(t, \tau)/dt = F\Psi(t, \tau) + L, \qquad \Psi(\tau, \tau) = 0. \tag{8.74}$$

The measurement equation is linearized about $\bar{x}(t_k)$ and a nominal value $\bar{p}$,

$$\delta y_{t_k} = M(t_k; \bar{x}(t_k), \bar{p})\,\delta x_{t_k} + N(t_k; \bar{x}(t_k), \bar{p})\,\delta p + v_k, \tag{8.75}$$

where

$$M(t_k; \bar{x}(t_k), \bar{p}) \triangleq \left[\frac{\partial h_i(\bar{x}(t_k), \bar{p}, t_k)}{\partial x_j}\right], \qquad N(t_k; \bar{x}(t_k), \bar{p}) \triangleq \left[\frac{\partial h_i(\bar{x}(t_k), \bar{p}, t_k)}{\partial p_j}\right].$$

The Schmidt–Kalman filter can now be directly applied to the system (8.73), (8.75). An extended Kalman–Schmidt filter (see Section 3, especially Theorem 8.1) can be readily written down. We leave these details for the reader.

Example 8.3. Show that

$$\Phi(t, \tau) = \left[\frac{\partial x_i(t)}{\partial x_j(\tau)}\right], \qquad \Psi(t, \tau) = \left[\frac{\partial x_i(t)}{\partial u_j(\tau)}\right]. \quad \blacktriangle$$

5. A SIMPLE EXAMPLE

A limited number of simple examples of linear filter applications are available in the literature. These include the analytical examples in Kalman [19], and a numerical example in Schmidt [31]. The scarcity of such examples is, of course, due to the nonlinearity of the covariance equations, which precludes closed-form solutions in all but the simplest cases. Since the detailed solution of a simple problem gives one a "feel" for the method, we think it useful to present some detailed computations and analyses of such a simple problem.

Our problem consists of a noise-free second-order system representing a falling body in a constant field

$$\ddot{z} = -g, \qquad t \geqslant 0, \tag{8.76}$$

z a scalar. Let the position be $z = (x)_1$, and the velocity $\dot{z} = (x)_2$. Then, defining the vector x,

$$x^{\mathrm{T}} = [(x)_1, (x)_2],$$

(8.76) can be written in system form

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ -g \end{bmatrix}, \qquad F \triangleq \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

It is easy to verify that the state transition matrix of this system is

$$\Phi(t, \tau) = \begin{bmatrix} 1 & t - \tau \\ 0 & 1 \end{bmatrix}, \tag{8.77}$$

so that

$$x_t = \Phi(t, \tau)x_\tau + \int_\tau^t \begin{bmatrix} 0 & t - s \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ -g \end{bmatrix} ds$$

$$= \Phi(t, \tau)x_\tau - g \begin{bmatrix} (t - \tau)^2/2 \\ (t - \tau) \end{bmatrix}.$$

Therefore,

$$x_{t+1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x_t - g \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}, \qquad t = 0, 1, \ldots . \tag{8.78}$$

We will take (scalar) observations of position

$$y_t = [1 \quad 0]\, x_t + v_t\,; \qquad t = 1, \ldots, 6; \qquad M \triangleq [1 \quad 0], \tag{8.79}$$

with $R(t) = 1$. We assume that $x_0 \sim N(\hat{x}_0, P_0)$,

$$\hat{x}_0 = \begin{bmatrix} 95 \\ 1 \end{bmatrix}, \qquad P_0 = \begin{bmatrix} 10 & 0 \\ 0 & 1 \end{bmatrix}.$$

Let us first determine whether our system (8.78), (8.79) is observable. The information matrix [from (7.151)],

$$\mathscr{I}(t, t-1) = \sum_{i=t-1}^{t} \Phi^{\mathrm{T}}(i, t)\, M^{\mathrm{T}}(i)\, R^{-1}(i)\, M(i)\, \Phi(i, t),$$

is computed to be

$$\mathscr{I}(t, t-1) = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}.$$

Therefore, our system is uniformly completely observable. [Take $\alpha = 0.1$, $\beta = 3$ in (7.153), for example.]

Take $g = 1$ in (8.76). Assume that the "true" initial condition is $(x_0)_1 = z(0) = 100$, $(x_0)_2 = \dot{z}(0) = 0$. Then the true trajectory (at $t = 1,\ldots, 6$) and a realization of the observations are given in Table 8.2. We show the details of one cycle of the filter computations as outlined in Section 2. The results for $t = 1,\ldots, 6$ are summarized in Table 8.2.

TABLE 8.2

FALLING BODY IN A CONSTANT FIELD

t	0	1	2	3	4	5	6
$(x_t)_1$	100.0	99.5	98.0	95.5	92.0	87.5	82.0
$(x_t)_2$	0	−1.0	−2.0	−3.0	−4.0	−5.0	−6.0
v_t		+0.5	−0.1	−1.1	+0.7	−0.2	+0.1
y_t		100.0	97.9	94.4	92.7	87.3	82.1
$\hat{x}_1(t \mid t)$	95.0	99.6	98.4	95.3	92.6	87.9	82.4
$\hat{x}_2(t \mid t)$	1.0	+0.36	−1.2	−2.3	−3.25	−4.6	−5.7
$P_{11}(t \mid t)$	10.0	0.88	0.66	0.66	0.62	0.56	0.50
$P_{22}(t \mid t)$	1.0	0.92	0.57	0.30	0.16	0.10	0.06

(Step 2)

$$\hat{x}(1 \mid 0) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 95 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.5 \\ 1.0 \end{bmatrix} = \begin{bmatrix} 95.5 \\ 0 \end{bmatrix},$$

(Step 3)

$$P(1 \mid 0) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 10 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 1 \\ 1 & 1 \end{bmatrix},$$

$$P(1 \mid 0)M^{\mathrm{T}} = \begin{bmatrix} 11 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 11 \\ 1 \end{bmatrix},$$

$$MP(1 \mid 0)M^{\mathrm{T}} = [1 \quad 0] \begin{bmatrix} 11 \\ 1 \end{bmatrix} = 11,$$

$$Y(1 \mid 0) = MP(1 \mid 0)M^{\mathrm{T}} + R = 12,$$

(Step 4)

$$K(1) = P(1 \mid 0)\, M^{\mathrm{T}} Y^{-1}(1 \mid 0) = \frac{1}{12}\begin{bmatrix}11\\1\end{bmatrix} = \begin{bmatrix}0.92\\0.08\end{bmatrix},$$

(Step 5)

$$\hat{x}(1 \mid 1) = \begin{bmatrix}95.5\\0\end{bmatrix} + \begin{bmatrix}0.92\\0.08\end{bmatrix}(100.0 - 95.5) = \begin{bmatrix}99.64\\0.36\end{bmatrix},$$

$$K(1)M = \begin{bmatrix}0.92\\0.08\end{bmatrix}[1 \quad 0] = \begin{bmatrix}0.92 & 0\\0.08 & 0\end{bmatrix},$$

$$I - K(1)M = \begin{bmatrix}0.08 & 0\\-0.08 & 1\end{bmatrix},$$

(Step 6)

$$P(1 \mid 1) = \begin{bmatrix}0.08 & 0\\-0.08 & 1\end{bmatrix}\begin{bmatrix}11 & 1\\1 & 1\end{bmatrix} = \begin{bmatrix}0.88 & 0.10\\0.10 & 0.92\end{bmatrix}.$$

Finally,

$$\hat{x}(6 \mid 6) = \begin{bmatrix}82.4\\-5.7\end{bmatrix}, \qquad P(6 \mid 6) = \begin{bmatrix}0.50 & 0.14\\0.14 & 0.06\end{bmatrix}.$$

The filtering results are also shown in Figs. 8.1 and 8.2. Plotted in

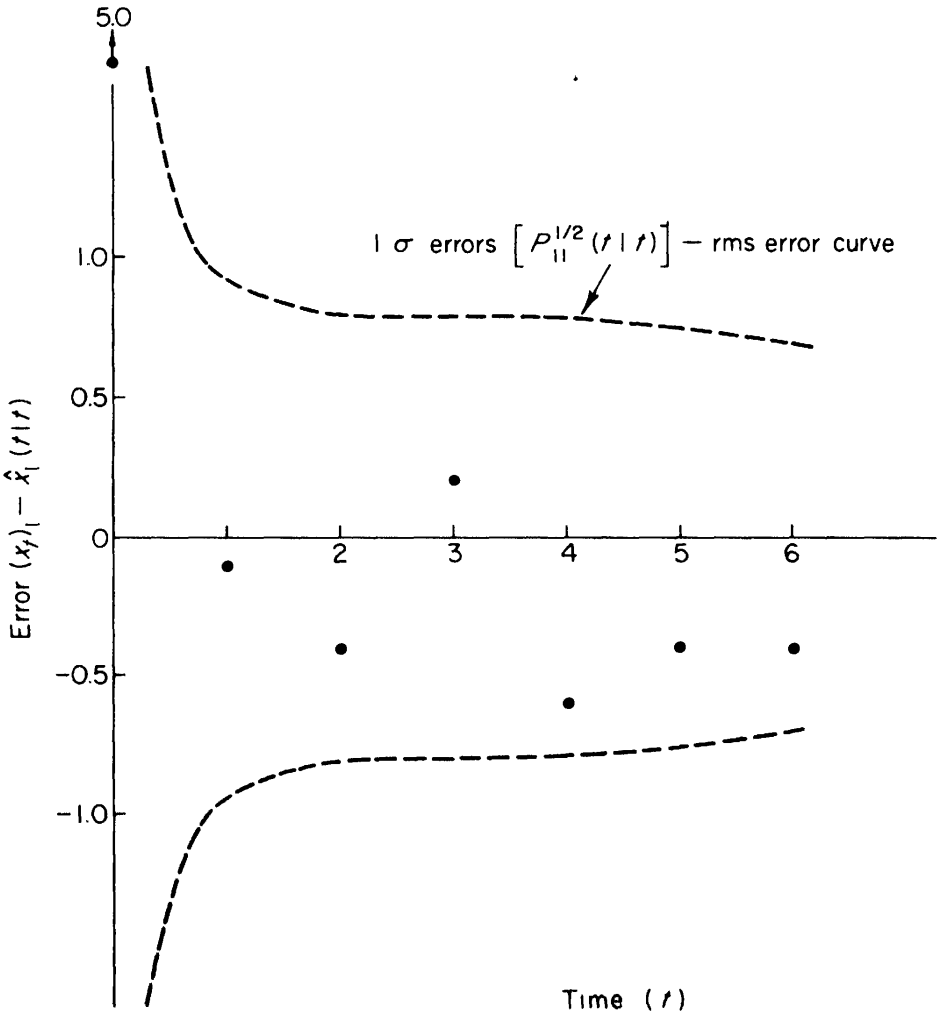


FIG. 8.1. Position estimation error.

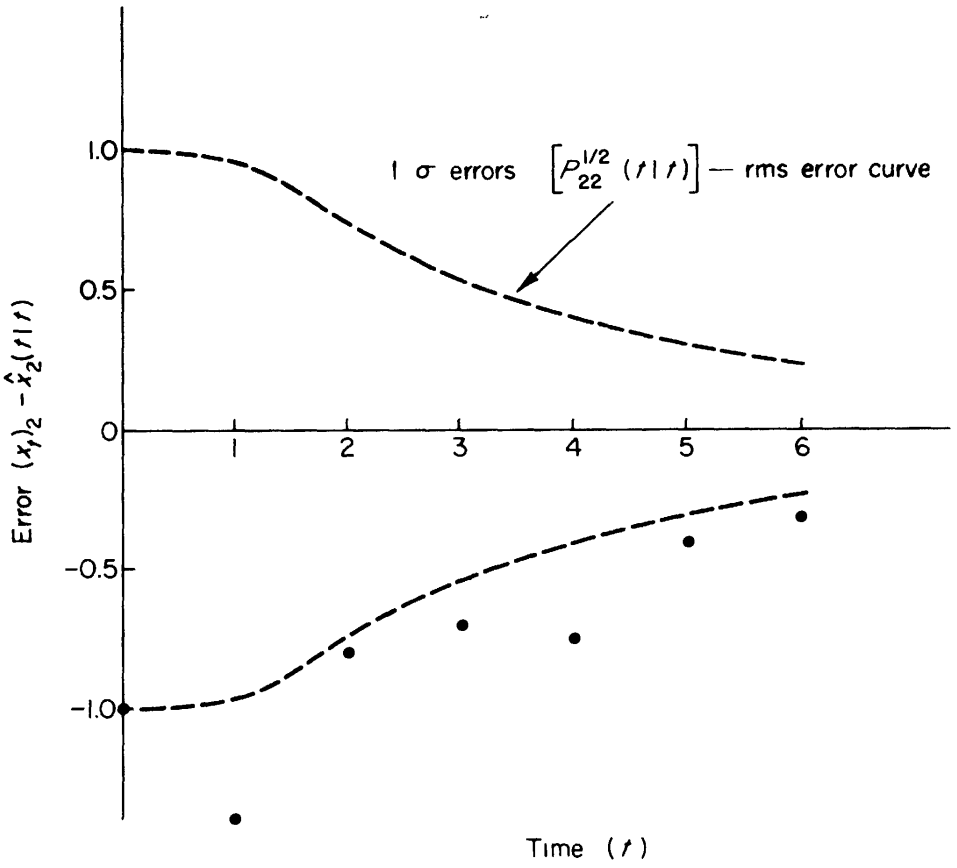


FIG. 8.2. Velocity estimation error.

these figures are the position and velocity estimation errors. The dashed lines represent the error standard deviations, or root mean square (rms) errors. If the filter were simulated many times, each time with different samples of x_0 and $\{v_t, t = 1,..., 6\}$, and if the estimation errors (dots) were plotted for each simulation, then about 67% of the dots would plot within the dashed lines. The fact that, in our particular simulation, the velocity estimation errors fall somewhat outside the rms curves is due to our particular choice of x_0 and the measurement noise samples. This initial transient effect would eventually disappear, and the appropriate proportion (67%) of the dots would then fall within the rms error curves. Recall that in this problem we are observing the position. As a result, the rms position error drops dramatically as soon as the first observation is processed. On the other hand, the rms velocity error does not decrease substantially until the second observation is processed. This is because two position observations are required to determine both components of the state vector. The dynamics of our system are such that velocity affects the position, whereas position does not affect velocity. As a consequence, velocity must first be estimated rather accurately before good estimates of position can be had. This explains the fact that after the second observation the velocity rms errors steadily

decrease, whereas position rms errors decrease much more slowly. Eventually, both rms errors will, of course, go to zero.

It is interesting to determine the effect of our initial uncertainty (P_0) on $P(6 \mid 6)$. This can be easily accomplished with the aid of the linear limited memory filter theory developed in Section 7.10. We first compute the predicted covariance matrix

$$P(6 \mid 0) = \Phi(6, 0)\, P_0 \Phi^{\mathrm{T}}(6, 0) = \begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 10 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 6 & 1 \end{bmatrix} = \begin{bmatrix} 46 & 6 \\ 6 & 1 \end{bmatrix},$$

and

$$P^{-1}(6 \mid 0) = \begin{bmatrix} 0.1 & -0.6 \\ -0.6 & 4.6 \end{bmatrix}.$$

Also,

$$P^{-1}(6 \mid 6) = \begin{bmatrix} 6.0 & -14 \\ -14 & 50 \end{bmatrix}.$$

Then, according to Theorem 7.10,

$$P^{-1}(6 \mid 6 \mid) = P^{-1}(6 \mid 6) - P^{-1}(6 \mid 0) = \begin{bmatrix} 5.9 & -13.4 \\ -13.4 & 45.4 \end{bmatrix},$$

and therefore

$$P(6 \mid 6 \mid) = \begin{bmatrix} 0.52 & 0.15 \\ 0.15 & 0.067 \end{bmatrix},$$

which is the covariance matrix of the error in the estimate conditioned on the observations *only*. Notice that $P(6 \mid 6 \mid)$ is only slightly "larger" than $P(6 \mid 6)$. That is, the prior data contain little information compared with the six observations.

What would the covariance matrix at $t = 6$ be had we used a different value of P_0, say P_0'? Take

$$P_0' = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}, \qquad P_0'^{-1} = \begin{bmatrix} 0.2 & 0 \\ 0 & 0.5 \end{bmatrix}.$$

Then

$$P'^{-1}(6 \mid 0) = \Phi^{-\mathrm{T}}(6, 0)\, P_0'^{-1} \Phi^{-1}(6, 0) = \begin{bmatrix} 0.2 & -1.2 \\ -1.2 & 7.7 \end{bmatrix}.$$

Then, according to Theorem 7.10,

$$P'^{-1}(6 \mid 6) = P^{-1}(6 \mid 6 \mid) + P'^{-1}(6 \mid 0) = \begin{bmatrix} 6.1 & -14.6 \\ -14.6 & 53.1 \end{bmatrix},$$

and

$$P'(6 \mid 6) = \begin{bmatrix} 0.48 & 0.13 \\ 0.13 & 0.055 \end{bmatrix}.$$

Our difference in initial conditions is

$$\delta P_0 = P_0 - P_0' = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix},$$

and

$$\delta P_6 = P(6 \mid 6) - P'(6 \mid 6) = \begin{bmatrix} 0.02 & 0.01 \\ 0.01 & 0.005 \end{bmatrix}.$$

If

$$\| A \| \triangleq \sum | a_{ij} |^2,$$

we see that

$$\| \delta P_0 \| = 26, \qquad \| \delta P_6 \| = 0.0006.$$

Thus we see that the effect of the initial condition on the estimation error is being washed out, as guaranteed by theory.

6. APPLICATIONS IN ORBIT DETERMINATION

We present in this section the results of some simulations of the extended Kalman filter (Theorem 8.1) in Earth satellite orbit determination. The reader unfamiliar with satellite orbit mechanics may refer to Appendix 8A. Simulated observations are used in these experiments. That is, a true orbit is assumed, observations are computed and contaminated with noise. As a consequence, the filter model is perfect, and none of the problems of divergence (Section 8) are encountered. Nevertheless, these simulations are indicative of results that can be obtained in real situations, and demonstrate the efficacy of the Kalman filter in determining the satellite orbit.[4]

Our example orbit determination problem takes place in two dimensions [(z, x) plane in Fig. 8A.2], and a nonrotating oblate Earth is used. The true orbit has the elements

$$a = 10{,}758.236 \quad \text{km (semimajor axis)},$$

$$e = 0.28427970 \quad \text{(eccentricity)},$$

$$\omega = -170.0 \quad \text{deg (argument of perigee)},$$

$$M = 5.0 \quad \text{deg (mean anomaly)}.$$

[4] The data presented in this section are due to Mr. R. K. Squires of the NASA Goddard Space Flight Center. We are indebted to Mr. Squires for making these data available.

Observations consist of range, range rate, and m direction cosines. The true orbit[5] and the observations are computed for 10.0 hours, or approximately three revolutions. The *a priori* estimates of the orbital elements were taken as

$$\hat{a} = 10{,}778.236 \quad \text{km},$$

$$\hat{e} = 0.28559740,$$

$$\hat{\omega} = -173.0 \quad \text{deg},$$

$$\hat{M} = 3.0 \quad \text{deg}.$$

The errors ($a - \hat{a}$, etc.) represent two standard deviations; that is, the initial covariance matrix P_0 is diagonal in the space of the orbital elements, with $(a - \hat{a})^2/4$, etc., along the diagonal.

The noise in the observations has the following standard deviations:

$$\sigma \text{ (noise in range)} = 0.2 \quad \text{km},$$

$$\sigma \text{ (noise in range rate)} = 1.0 \quad \text{m/sec},$$

$$\sigma \text{ (noise in } m \text{ direction cosine)} = 0.005.$$

Three tracking stations, spaced 120 deg apart, observe the satellite. Station 1 sees the satellite for $\sim$0.6 hr, station 2 for $\sim$0.4 hr, and station 3 for $\sim$1.4 hr in each revolution. The standard data rate is one observation every $^1/_{32}$ hr, and the high data rate is one observation every $^1/_{512}$ hr ($\sim$7 sec). An observation may be a scalar (e.g., range) or a vector (e.g., range and range rate), as will be indicated.

Figure 8.3 shows the periodic variations in the semimajor axis caused by earth oblateness (see Appendix 8A). The oblateness causes a to depart from its constant (Keplerian) value; the element a is said to be *osculating*. This figure is presented here only for its general interest.

A comparison of the effectiveness of the various observation types in determining the orbit is given in Figs. 8.4 and 8.5. The magnitude of the actual estimation errors in the semimajor axis is plotted in Fig. 8.4. We see that range observations determine the orbit best. Note that, for range observations, the actual error in a is of the order of 10 m after one revolution, even though the noise in the range observations has a standard deviation of 200 m. We are indeed "filtering out" the noise. The rms errors in a are plotted in Fig. 8.5. These are the standard

[5] The filter computations were actually performed using the NASA variables [24] as state variables. These have certain advantages over the Cartesian position and velocity coordinates.

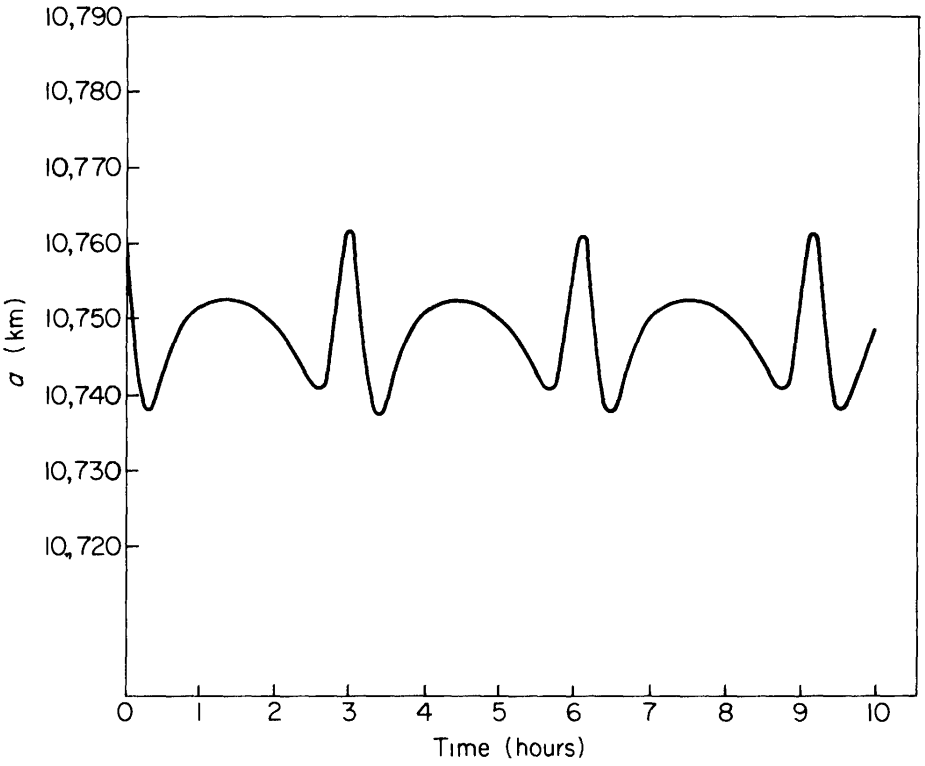


FIG. 8.3. Osculating semimajor axis.

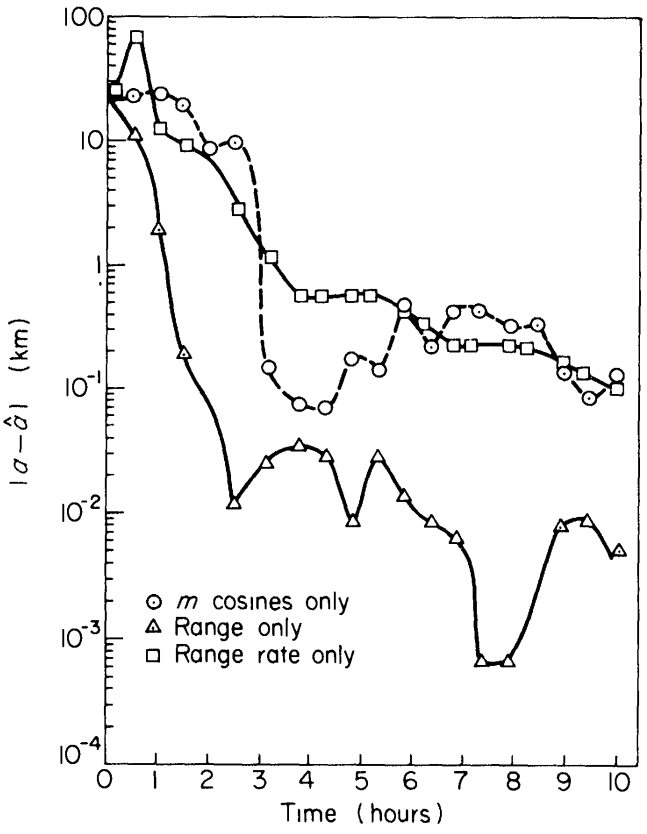


FIG. 8.4. Estimation errors in a.

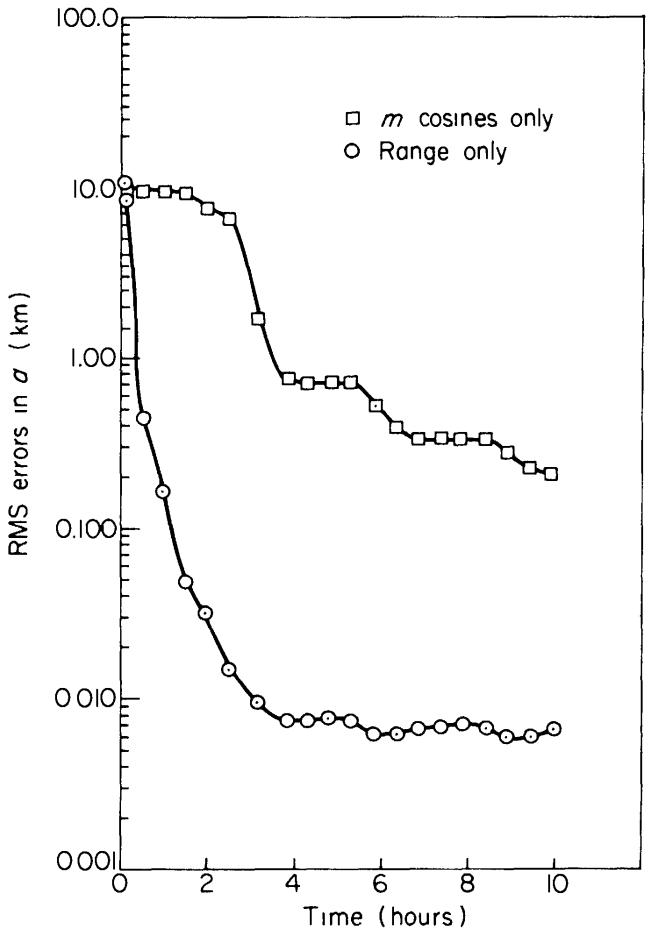


FIG. 8.5. *RMS* errors in *a*.

deviations as predicted by the error covariance matrix. We see that the actual errors are consistent with their statistics. That is, the error covariance matrix faithfully represents the variance of the errors, as in theory it should.

The rate at which observations are taken has a profound effect on orbit estimation accuracy, especially if the observations are very noisy. This is illustrated in Fig. 8.6, where the *m* direction cosines were used in the estimation. Notice that, using a single station and the standard data rate, virtually no improvement in the relative position error is obtained. One can, however, determine the position quite accurately using all three stations and the high data rate. We have determined the position to about 10 m even with the very noisy *m* direction cosines.

In actual practice, a tracking station takes several observation types simultaneously. The effect of multiple data (at the same time instant) is shown in Fig. 8.7. We see that *m* direction cosines add little or nothing to the information contained in the range data. The addition of range rate observations improves the estimate somewhat, but not in any profound way. One should not attempt to generalize on the basis of Fig. 8.7, however, as station location, the number of stations, etc., have

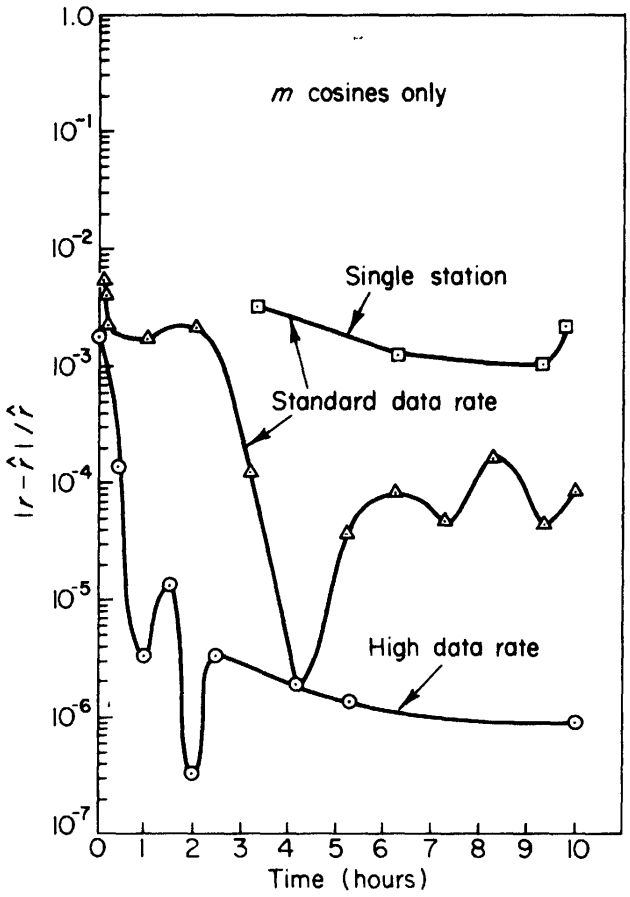


FIG. 8.6. Relative errors in position magnitude.

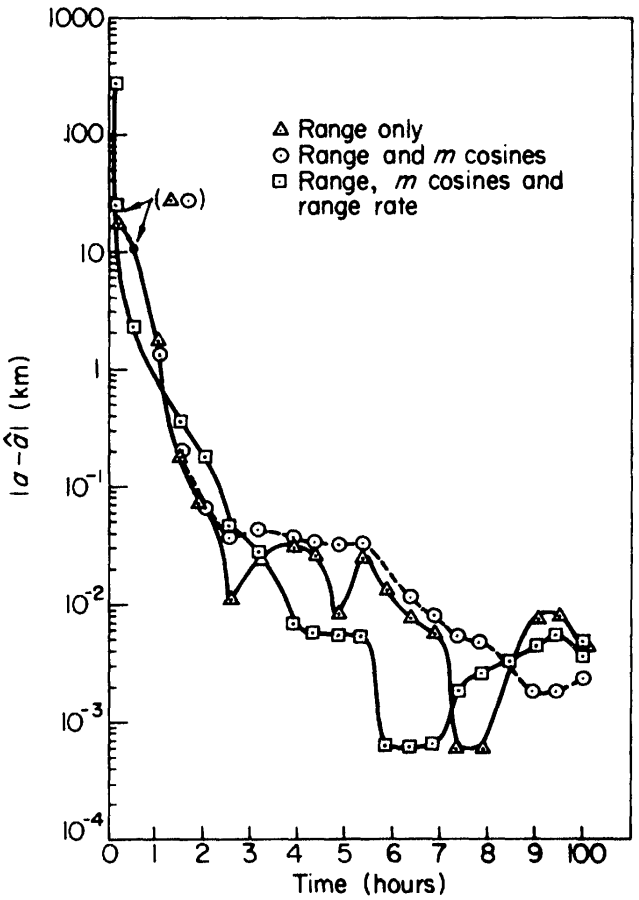


FIG. 8.7. Effect of multiple data types.

some effect on the value of multiple data. Generally speaking, the m cosines are so noisy that they do not enhance an estimate based on range data alone. Range rate observations can, however, substantially improve an estimate based on range alone.

The results of this section are indicative of what might be obtained in real satellite orbit determination. The extended Kalman filter promises to be a valuable tool in this field. As has been noted in the Introduction, the Kalman filter is already in use in several aerospace applications. As was also noted, the modeling problem still remains. We pursue that problem in Section 8.

7. APPLICATIONS IN REENTRY

Reentry trajectory estimation differs from orbit determination in several significant ways. For one thing, the dominant forces in reentry are aerodynamic rather than gravitational. Because of our relatively poor knowledge of the atmosphere (density, temperature, winds, etc.), these forces are not well known. As a consequence, the modeling problem is much more severe in reentry than it is in satellite orbit determination. Furthermore, reentry dynamics are much more nonlinear than orbit dynamics (see Appendix 8B). On the other hand, reentry trajectories are of much shorter time duration than (several) satellite orbit revolutions, so that in reentry, model errors propagate over shorter time arcs. We shall not attempt to determine which of these two estimation problems is the more difficult. We can, however, expect that nonlinearities will have a more significant effect in reentry.

Through the courtesy of Mr. William E. Wagner of the Martin Company, and by permission of the *Journal of Spacecraft and Rockets* (*AIAA*), we reproduce here some reentry estimation experiments performed by Mr. Wagner and reported by him [39]. These are simulated experiments; that is, the data are generated, not real. Nevertheless, these experiments are instructive, and indicate the kind of results which can be expected in applying the Kalman filter to reentry trajectory estimation. The reader is referred to Appendix 8B and to Wagner's work [39] for a description of reentry mechanics.

The "true" reentry initial conditions (at time $t = 0$) in these experiments are

$$v = 25{,}093 \text{ fps}, \qquad h = 400{,}000 \text{ ft},$$

$$\gamma = -2.0422 \text{ deg}, \qquad \varphi = 24.8 \text{ deg},$$

$$\psi = -168.348 \text{ deg}, \qquad \theta = 215.25 \text{ deg}.$$

The prior estimates of these state variables are given in the second line of Table 8.3. Observations consist of ground-based radar tracking data (range, azimuth, and elevation) from three stations, and telemetered onboard acceleration measurements. Onboard accelerations are measured in two orthogonal directions. The three tracking zones and (line-of-sight) elevation angles are shown in Fig. 8.8. The (standard) noise in the observations has the following standard deviations:

$$3\sigma \text{ (range)} = 270 \text{ ft},$$

$$3\sigma \text{ (azimuth)} = 4.05 \text{ deg},$$

$$3\sigma \text{ (elevation)} = 4.23 \text{ deg},$$

$$3\sigma \ (x\text{-acceleration}) = 3\%,$$

$$3\sigma \ (z\text{-acceleration}) = 3\%.$$

The standard data rate is one set of these observations every 4 sec.

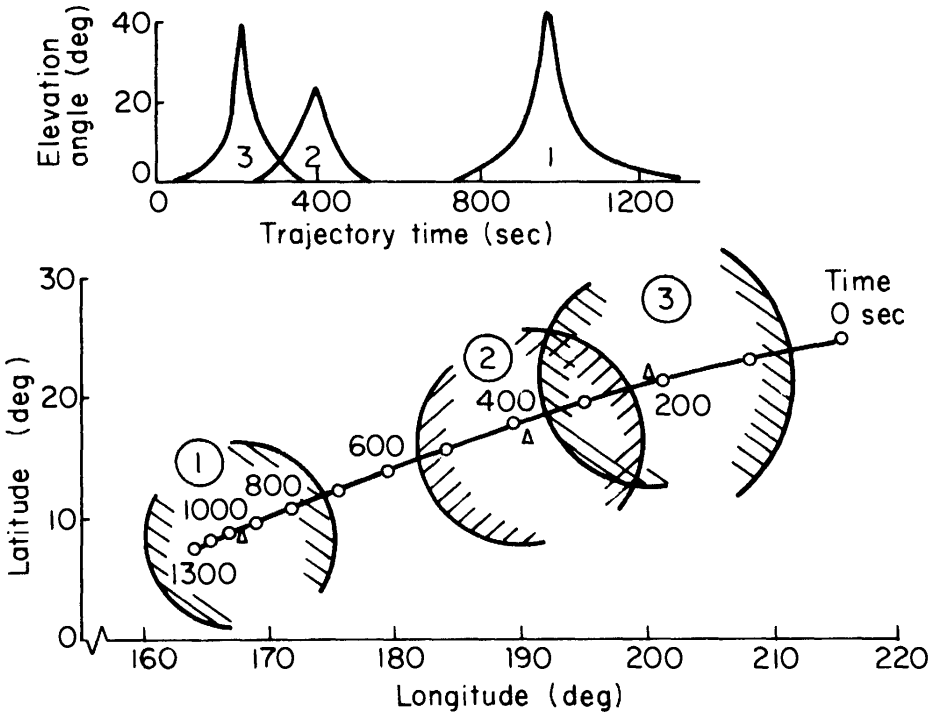


FIG. 8.8. Tracking acquisition zones.

The linear filter is applied to this reentry estimation in two ways. In one case, a single reference trajectory is used as described in Section 3. In the second case, the extended Kalman filter (Theorem 8.1) is used. In each case, the first linearization is about the prior values (estimates). As is seen in Table 8.3, the prior estimation errors are rather large. The estimation results are displayed in Table 8.3 in terms of the estimates of (and estimation errors in) initial conditions (at time $t = 0$).

TABLE 8.3

INITIAL CONDITION ESTIMATION RESULTS

Iter.	v	$-\gamma$	$-\psi$	h	ϕ	θ	Δv	$\Delta\gamma$	$\Delta\psi$	Δh	$\Delta\phi$	$\Delta\theta$
true	25093.608	2.0422383	168.34792	400000.00	24.800000	215.25197	—	—	—	—	—	—
prior	24593.607	5.0422383	166.34791	420000.00	25.400000	215.85196	−500.	−3.00	2.00	20000.	0.600	0.600
					Global iterations of extended Kalman filter							
1	25107.221	2.0996928	168.34696	402809.67	24.798588	215.24566	13.6	−0.057	0.001	2809.	−0.001	−0.006
2	25098.217	2.0537964	168.34372	400058.93	24.801576	215.24589	4.61	−0.012	0.004	58.9	0.002	−0.006
3	25098.097	2.0533670	168.34369	400039.77	24.801577	215.24579	4.49	−0.011	0.004	39.8	0.002	−0.006
					Global iterations using single reference each iteration							
2	25098.217	2.0537964	168.34372	400058.93	24.801576	215.24589	4.61	−0.012	0.004	58.93	0.002	−0.006
3	25080.183	2.0014297	168.34544	398775.88	24.801116	215.24873	−13.43	0.0408	0.002	−1224.1	0.001	−0.003
4	25088.111	2.0266663	168.34500	399610.04	24.801433	215.24854	−5.50	0.0156	0.003	−390.0	0.001	−0.003
5	25089.525	2.0308211	168.34487	399713.25	24.801435	215.24817	−4.08	0.0114	0.003	−286.8	0.001	−0.004

That is, after all the data are processed, the estimate is smoothed back to time zero.

Wagner found that the filter that used a single reference trajectory diverged and was unusable. This is due to the large initial estimation errors and the nonlinearities in the problem. The extended Kalman filter performed relatively well. Its results are summarized in line 1 of Table 8.3. The errors in all the state components, except altitude, are relatively small. The altitude error is, however, unsatisfactory.

Assuming that the unsatisfactory altitude error is caused by significant nonlinearities neglected in the filter, the extended Kalman filter was iterated globally, as described in Section 3. Three iterations are summarized in Table 8.3. It is seen that these iterations lead to very satisfactory results, inasmuch as the estimation errors are rather small.

Using the results of the extended Kalman filter as the first iteration, the single reference filter was iterated several times. The results of these iterations are also given in Table 8.3. It is seen that altitude errors are quite large even after five iterations. This demonstrates the advantage of the relinearization procedure used in the extended Kalman filter. These results also show that nonlinear effects are quite significant in reentry problems.

Timewise convergence of the estimates is shown (for altitude) in Fig. 8.9. We see that the true altitude is acquired very rapidly, and that the estimate subsequently oscillates about the true value with ever-decreasing error amplitude.

An error analysis for this reentry problem is given in Fig. 8.10.

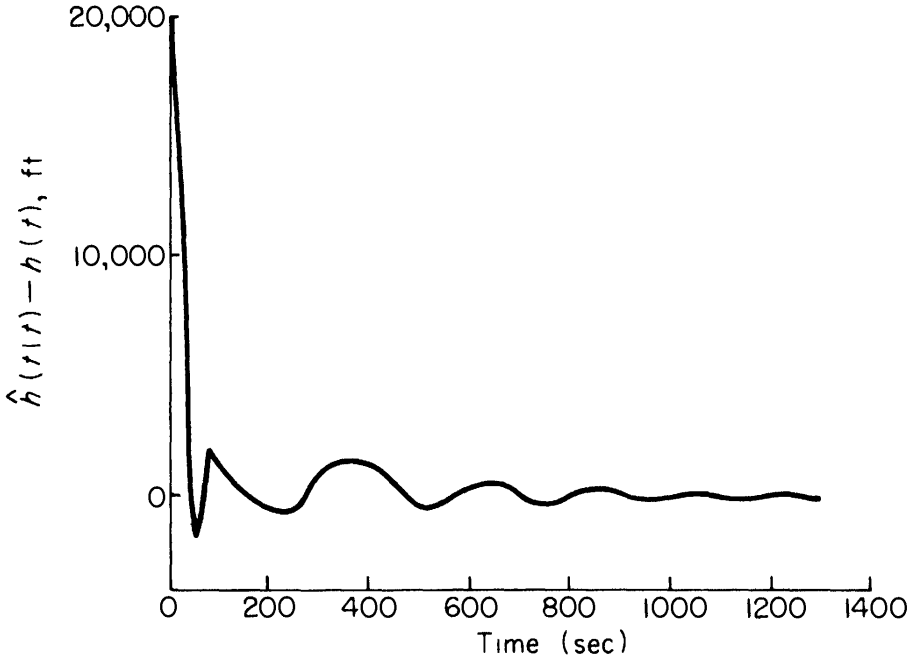


FIG. 8.9. Timewise convergence.

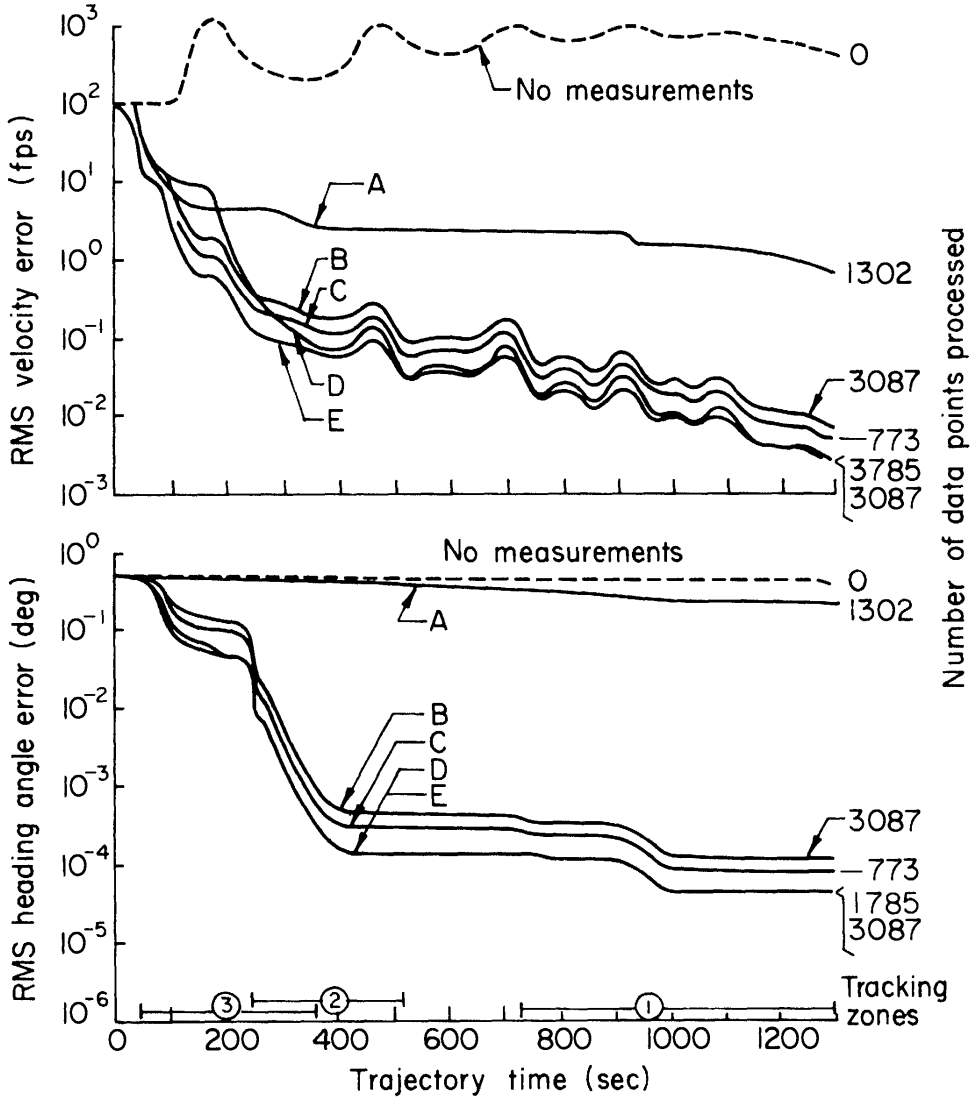


FIG. 8.10. Standard deviations of Errors in v and ψ. A: telemetry, 2-sec data rate; B: tracking and telemetry, 2-sec data rate; C: tracking and telemetry, 8-sec data rate; D: tracking, 2-sec data rate; E: tracking and telemetry, 2-sec data rate (A–E, all standard measurement noise).

Plotted are the rms errors in velocity and heading angle (square roots of the appropriate elements of the error covariance matrix P). Figure 8.10 shows the effect of using tracking and/or telemetered acceleration data, measurement noise, and data rate. We see that tracking data are of greater value in estimating the trajectory than are the acceleration data. The fluctuations in the rms velocity error are caused by the skipping motion of the reentry vehicle.

8. FILTER DIVERGENCE AND ERROR COMPENSATION TECHNIQUES

The problem of filter divergence was already noted in Sections 6.11, 7.8, and elsewhere. In essence, the problem is this. If the filter is

constructed on the basis of an erroneous model, it "learns the wrong state too well" when it operates over many data. The problem is particularly acute when the noise inputs to the system are small and when measurement noise is small. For then the filter is capable of learning the state very well. Eventually, the error covariance matrix becomes very small, the filter gain is therefore small, and subsequent observations have little effect on the estimate. But the dynamical system model in the filter is different from the actual system model, so that the estimate and the state can diverge. In actual applications, the onset of divergence manifests itself by the inconsistency of residuals [e.g., Eq. (8.8)] with their predicted statistics. Residuals become biased [nonzero mean (8.9)] and larger in magnitude than their rms values, as predicted by (8.10).

Divergence problems have been particularly acute in orbit determination [41]. This is because measurement devices (radars) are very accurate, many data are available, and the filter is required to operate over many revolutions of the satellite orbit. As we noted in Section 1, the modeling or divergence problem remains a serious problem in orbit determination.

To give the reader a "feel" for the divergence problem, we present a simple analytical example[6] of divergence borrowed from Schlee *et al.* [29]. Consider the problem of estimating the altitude h of a space vehicle from altimeter data. Suppose the filter is designed assuming that the altitude is a constant, whereas in actual fact the vehicle is climbing at a constant rate. The observations are

$$y_k = h_k + v_k, \tag{8.80}$$

and, in the notation of Section 7.8,

$$\bar{h}_{k+1} = \bar{h}_k. \tag{8.81}$$

Our filter design is [see (7.206)–(7.208)]

$$\hat{h}_k^k = \hat{h}_{k-1}^{k-1} + \frac{P_c(k-1 \mid k-1)}{P_c(k-1 \mid k-1) + R}(y_k - \hat{h}_{k-1}^{k-1}), \tag{8.82}$$

where the recursion for the computed error variance is

$$P_c(k \mid k) = P_c(k-1 \mid k-1) - \frac{P_c^2(k-1 \mid k-1)}{P_c(k-1 \mid k-1) + R}. \tag{8.83}$$

[6] The paper by Schlee *et al.* [29] contains other examples of filter divergence in orbit determination. Additional examples may also be found in Fitzgerald [13] and in Sections 11 and 12.

Equation (8.83) has the solution

$$P_c(k \mid k) = \frac{P_0 R}{kP_0 + R}, \tag{8.84}$$

so that (8.82) becomes

$$\hat{h}_{k+1}^{k+1} = \hat{h}_k^k + \frac{P_0}{(k+1)P_0 + R}(y_{k+1} - \hat{h}_k^k). \tag{8.85}$$

Since, in actuality,

$$h_k = h_0 + ks \qquad (h_{k+1} = h_k + s) \tag{8.86}$$

(the vehicle is climbing at a constant speed), the recursion for the actual estimation error

$$\tilde{h}_k^k \triangleq h_k - \hat{h}_k^k$$

is

$$\tilde{h}_{k+1}^{k+1} = [\tilde{h}_k^k + s]\left[\frac{kP_0 + R}{(k+1)P_0 + R}\right] - \frac{P_0 v_{k+1}}{(k+1)P_0 + R}, \tag{8.87}$$

which has the solution

$$\tilde{h}_k^k = \frac{R}{kP_0 + R}\tilde{h}_0^0 + \frac{[(k-1)k/2]P_0 + kR}{kP_0 + R}s - \frac{P_0}{kP_0 + R}\sum_{i=1}^{k} v_i. \tag{8.88}$$

Now the first term in (8.88) goes to zero as $k \to \infty$. The last term also goes to zero in the mean square sense, say (the familiar "square root of k" law). However, the middle term involving the speed s is unbounded in k and goes to infinity as $(k - 1)/2$. As we see, the estimation errors are unbounded; the filter is diverging. Note, however, that, if s were zero, we could learn the altitude arbitrarily well given sufficient observations.

Example 8.4. Use the techniques of Section 7.8 to analyze the effect of the model error in this example. The actual system can be considered to be the second-order system

$$h_{k+1} = h_k + s_k, \qquad s_{k+1} = s_k,$$

with the state transition matrix

$$\Phi(k+1, k) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$$

whereas the erroneous model has the transition matrix

$$\Phi_e(k+1, k) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The model error can thus be viewed as an error in Φ. Alternatively, the model error can be viewed as the error $\Delta\varphi = s$ in the first-order system (8.86). ▲

Schlee *et al.* [29] show that actual estimation errors can be bounded (in this example) by adding noise to the system (8.81):

$$\bar{h}_{k+1} = \bar{h}_k + \bar{w}_k .$$

This is, in fact, a special case of the corollary to Theorem 7.7. Identify s in (8.86) with $\Delta\varphi(k)$ in Section 7.8. We see that the addition of system noise gives us uniform complete controllability. The other hypotheses of the corollary are clearly satisfied and, as a consequence, estimation errors are bounded. The important question is, of course, how much noise to add. Furthermore, it is one thing to bound the errors and quite another to assure that the resulting filter performance is satisfactory. The "bounded" errors may be too large for the results to be usable.

The error analyses of Section 7.8 provide us with numerical tools for evaluating the effect of model errors on filter performance. As was mentioned there, such errors may be intentional. That is, certain approximations might be made to simplify the filter computations. On the other hand, model errors may be present because of our limited knowledge of the physical system. Furthermore, computational errors resulting from the use of finite precision arithmetic effectively produce errors in the model. In any case, the tools and qualitative results developed in Section 7.8 are not enough. Faced with model errors, we require techniques for synthesizing a filter that works. That is, a filter that not only produces bounded errors, but also one that gives satisfactory performance. The actual error variance should be sufficiently small, and the estimation errors themselves should be consistent with their predicted statistics.

As was already noted, divergence takes place when the covariance matrix becomes too small or optimistic. For then the filter gain K (8.4) becomes small, and subsequent observations are ignored. In the example of divergence discussed previously, we see from (8.84) that

$$P_c(k \mid k) \to 0 \qquad (k \to \infty),$$

and consequently

$$K_c(k) = \frac{P_c(k-1 \mid k-1)}{P_c(k-1 \mid k-1) + R} \to 0 \qquad (k \to \infty).$$

Then the estimate evolves according to (8.81), whereas the actual state evolves according to (8.86), and divergence takes place. The basic idea,

then, in order to prevent divergence, is to increase the covariance matrix P. This is justified, since such an increase in the covariance matrix *compensates* for the model errors. Model errors in some sense add uncertainty to the system which should reflect itself in some degradation in certainty (increase in P).

Two somewhat different points of view can be taken with respect to model error compensation. One is to provide a (fictitious) noise input to the system or to attribute the errors to inaccuracies (uncertainties) in some of the parameters of the system, thus effecting an increase in the covariance matrix directly. The other point of view is to overweight the most recent data in some way. This causes the filter to forget old information, and thus indirectly increases the covariance matrix. The physical justification for forgetting old observations is that old observations, when predicted over long time arcs through an erroneous system, can be valueless. We saw that the first approach of adding a fictitious noise input is effective in the example described in this section. Recalling, however, the discussion of Theorem 7.7, we note that fictitious noise may not always be effective in preventing divergence.

The next four sections are devoted to the development of these two points of view in model error compensation. Several error compensation techniques are described and discussed. Sections 11 and 12 describe more sophisticated approaches to error compensation, together with some interesting applications in orbit determination.

9. FICTITIOUS NOISE INPUTS AND PARAMETER UNCERTAINTIES

It is possible to parametrize those portions of the dynamics (and/or measurement functions) which are poorly known, and consider the parameters in such a representation as unknown or poorly known quantities. Any well-behaved function can be approximated arbitrarily well by polynomials or by other series. One could then augment the state vector with these parameters and estimate them together with the state, as described in Section 4. This approach may, of course, lead to a high-dimensional state vector which, in turn, greatly increases the filter computational load. Furthermore, unless a sufficient number of parameters are provided, we may be simply postponing the inevitable. Once we learn the wrong parametric representation, the filter can again diverge. On the other hand, if too many parameters are used, there may be insufficient data for filtering. As a rather absurd extreme, we could imagine fitting an nth-order polynomial to $n + 1$ data points. In

this connection, the reader is referred to an interesting nonparametric approach of Dennis [9].

Instead of estimating these uncertain parameters, we could, via the Schmidt–Kalman filter (Theorem 8.4), include their uncertainty in the state filter. The parameter uncertainty will always degrade the state estimate, since we are not improving our estimate of these parameters themselves. This is a desirable feature that will tend to keep the covariance matrix open. The Schmidt–Kalman filter has, for example, been successfully used by Wolf [41] for drag and earth-mass uncertainties in earth satellite orbit determination. It was found, however, that uncertain parameters, which lead to successful orbit determination for one satellite, do not produce good results in the case of another satellite. Obviously, different parameters were dominant in each case. Which parameters to use in a given problem has to be determined by experimentation.

As we saw in the preceding section, fictitious noise inputs, which essentially cover (deterministic) model errors, can be effective in preventing divergence. In taking this approach to divergence, one has to determine what noise to add. This amounts to specifying $\Gamma Q \Gamma^{\mathrm{T}}$ in (8.3). An effective $\Gamma Q \Gamma^{\mathrm{T}}$ can be determined by trial and error, and this has been successfully done by Fitzgerald [13], for example. One simulates the filter for various values of $\Gamma Q \Gamma^{\mathrm{T}}$ until one gets satisfactory filter performance.

Wolf [41] has used more systematic approaches to determine the noise input. One such approach is to determine the major "direction(s)" in state space which are affected by model errors. For example, errors in drag affect in-track satellite motion. This essentially defines the noise coefficient matrix Γ, and one can set

$$\Gamma Q \Gamma^{\mathrm{T}} = k^2 \, \Delta t^2 \, \Gamma \Gamma^{\mathrm{T}}, \tag{8.89}$$

where Δt is the time interval between observations and the parameter k is to be determined by experimentation. Another approach is to pick (perhaps by experimentation) a likely parameter error Δa, say, and, using the dynamical equations, actually compute $\Delta x_{t_{k+1}}$ due to Δa at t_k. Then set

$$\Gamma(t_k) \, Q(t_{k+1}) \, \Gamma^{\mathrm{T}}(t_k) = \Delta x_{t_{k+1}} \, \Delta x_{t_{k+1}}^{\mathrm{T}} . \tag{8.90}$$

Yet another approach due to Wolf [41] is the following. Assume a likely noise level due to errors in the system. This would be noise in the accelerations in satellite orbit problems. Generate a random sample of this noise and pass it through the dynamical system (from t_k to t_{k+1}). The noise sample produces a $\Delta x_{t_{k+1}}$, which is used in (8.90). These techniques take into account the effect that model errors have on the

system, and are aids in determining appropriate fictitious noise inputs.

Pines *et al.* [24] have developed a computational (machine) error noise model that is useful when the computations are performed with limited (e.g., single) precision. Suppose the computer has a p-digit word length. Then a number x in the computer is of the form

$$x = .\, x_1 x_2 \cdots x_p 10^n.$$

Assume that the computer simply drops the number beyond the least significant digit in addition. Then the addition round-off error is of the same sign as x, and has magnitude 10^{n-p},

$$v_x \sim \frac{x}{|x|} 10^{n-p}.$$

Taking $|x| \sim 10^n$, we have

$$v_x \sim x10^{-p}.$$

Assuming that this round-off takes place in the prediction (8.2), we use the noise covariance matrix

$$\Gamma(t_k)\, Q(t_{k+1})\, \Gamma^{\mathrm{T}}(t_k) = 10^{-2p} \begin{bmatrix} \hat{x}_1^2(t_{k+1} \mid t_k) & & 0 \\ & \ddots & \\ 0 & & \hat{x}_n^2(t_{k+1} \mid t_k) \end{bmatrix}, \quad (8.91)$$

and assume that round-off errors are uncorrelated.

This computational error model is useful when model errors are in the machine noise level. Schlee *et al.* [29] have used it successfully in cases where model errors are larger than computational errors, by simply making p smaller. They used single precision arithmetic ($p = 8$) and engineered a $p = 6.5$.

The proliferation of so many techniques of adding uncertainty to the system to account for model errors is evidence of the fact that none of them work very well. All these techniques have their "fiddle parameters," which have to be adjusted in each application by experimentation. As a consequence, these techniques are not applicable in real-time filtering. In Section 11, we describe an *adaptive* technique of estimating noise input variances in real time. We let the actual filter measurement residuals determine the noise inputs for us.

10. OVERWEIGHTING THE MOST RECENT DATA

As we noted in Section 8, an indirect approach to increasing the covariance matrix, and thus increasing the filter gain, is to degrade old

data (observations) or, equivalently, to overweight more recent observations. This has the effect of decreasing the information (matrix) on which the estimate is based. (Recall that error variance is inversely proportional to information.) The reasoning behind this approach is that model errors themselves degrade the value of information in the distant past. We now describe several techniques for this type of model error compensation.

Fagin [10] describes the following technique of *exponentially age-weighting* old data. Consider the recursive least squares derivation of the linear filter in Example 7.2. In the minimization of J_k (7.37), replace

$$R_1 , R_2 ,..., R_k$$

by

$$e^{(t_k-t_1)/\tau}R_1 , e^{(t_k-t_2)/\tau}R_2 ,..., e^{(t_k-t_k)/\tau}R_k , \qquad \tau > 0.$$

In the minimization of J_{k+1}, replace

$$R_i , \qquad i = 1,..., k$$

by

$$e^{(t_{k+1}-t_k)/\tau}R_i , \qquad i = 1,..., k.$$

This leads to a filter with gain matrix

$$K_e(k+1) = P(k+1 \mid k)\, M^{\mathrm{T}}(k+1) \times [M(k+1)\, P(k+1 \mid k)\, M^{\mathrm{T}}(k+1) + e^{-(t_{k+1}-t_k)/\tau}R(k+1)]^{-1}, \tag{8.92}$$

and the recursion for P (at an observation)

$$P(k+1 \mid k+1) = e^{(t_{k+1}-t_k)/\tau}[I - K_e(k+1)\, M(k+1)]\, P(k+1 \mid k). \tag{8.93}$$

The exponentially age-weighted filter then consists of the filter described in Section 2 [(8.2)–(8.6)] with (8.4) replaced by (8.92), and (8.6) replaced by (8.93). Note that, the smaller τ is, the faster old observations are "forgotten."

A simple means of keeping the filter gain from becoming small is to fix the covariance matrix P at some value, say $\bar{P}$. $\bar{P}$ could be engineered *a priori*, or could be the result of filtering a given amount of data. The filter gain would then be

$$K_f(k+1) = \bar{P}M^{\mathrm{T}}(k+1)[M(k+1)\, \bar{P}M^{\mathrm{T}}(k+1) + R(k+1)]^{-1}, \tag{8.94}$$

and

$$\hat{x}(k+1 \mid k+1) = \hat{x}(k+1 \mid k) + K_f(k+1)[y_{k+1} - M(k+1)\, \hat{x}(k+1 \mid k)]. \tag{8.95}$$

The recursion for the estimation error covariance matrix for this filter is easily computed to be

$$P(k+1 \mid k+1) = [I - K_f(k+1)\, M(k+1)] \times P(k+1 \mid k)[I - K_f(k+1)\, M(k+1)]^{\mathrm{T}} + K_f(k+1)\, R(k+1)\, K_f^{\mathrm{T}}(k+1). \quad (8.96)$$

This technique was successfully applied by Fitzgerald [13]. It appears quite attractive for a real-time filtering situation that can be extensively simulated to arrive at a design for $\bar{P}$.

Schmidt [32, 33] has proposed several techniques of overweighting current observations. They are applicable to *scalar* observations; vector observations can usually be filtered by processing their components one at a time. One of these techniques, which appears to be the most interesting, is described next.

Schmidt's technique consists of computing an estimate that is a linear combination of the estimate based on the current observation and past data, and the estimate based on the current observation alone. Let

$$\Delta x \triangleq x(k+1) - \hat{x}(k+1 \mid k), \quad (8.97)$$

$$\Delta \hat{x} \triangleq \hat{x}(k+1 \mid k+1) - \hat{x}(k+1 \mid k), \quad (8.98)$$

$$\Delta y \triangleq y_{k+1} - M(k+1)\, \hat{x}(k+1 \mid k) = M(k+1)\, \Delta x + v_{k+1}. \quad (8.99)$$

The estimate based on the current observation and past data is the usual Kalman estimate

$$\Delta \hat{x} = P(k+1 \mid k)\, M^{\mathrm{T}}(k+1) \times [M(k+1)\, P(k+1 \mid k)\, M^{\mathrm{T}}(k+1) + R(k+1)]^{-1}\, \Delta y. \quad (8.100)$$

Note that the quantity in brackets is a scalar, since observations are scalar. Let us determine the estimate of Δx based on y_{k+1} only. From (8.99), the least squares estimate is

$$\Delta \hat{x} = [M^{\mathrm{T}}(k+1)\, R^{-1}(k+1)\, M(k+1)]^{\#}\, M^{\mathrm{T}}(k+1)\, R^{-1}(k+1)\, \Delta y, \quad (8.101)$$

where $A^{\#}$ is the pseudo-inverse of A. But

$$M^{\#} = M^{\mathrm{T}}/MM^{\mathrm{T}},$$

since M is a row matrix, so that (8.101) becomes

$$\Delta \hat{x} = \frac{M^{\mathrm{T}}(k+1)}{M(k+1)\, M^{\mathrm{T}}(k+1)}\, \Delta y,$$

or

$$\Delta\hat{x} = cM^{T}(k+1)\,R(k+1)[M(k+1)\,M^{T}(k+1)]^{-1} \times [M(k+1)\,P(k+1\mid k)\,M^{T}(k+1) + R(k+1)]^{-1}\,\Delta y, \quad (8.102)$$

with the appropriate constant c. Taking our estimate to be a linear combination of (8.100) and (8.102), we have the estimator

$$\begin{aligned}\hat{x}(k+1\mid k+1) = \hat{x}(k+1\mid k) &+ [P(k+1\mid k)\,M^{T}(k+1) \\ &+ cM^{T}(k+1)\,R(k+1)[M(k+1)\,M^{T}(k+1)]^{-1}] \\ &\times [M(k+1)\,P(k+1\mid k)\,M^{T}(k+1) + R(k+1)]^{-1} \\ &\times [y_{k+1} - M(k+1)\,\hat{x}(k+1\mid k)]. \end{aligned} \quad (8.103)$$

Multiplying (8.103) on the left by M, we have

$$\begin{aligned} &M(k+1)\,\hat{x}(k+1\mid k+1) \\ &\quad = M(k+1)\,\hat{x}(k+1\mid k) \\ &\qquad + \frac{[M(k+1)\,P(k+1\mid k)\,M^{T}(k+1) + cR(k+1)]}{[M(k+1)\,P(k+1\mid k)\,M^{T}(k+1) + R(k+1)]}\,\Delta y. \end{aligned}$$

Thus, with $c = 1$,

$$M(k+1)\,\hat{x}(k+1\mid k+1) = M(k+1)\,\hat{x}(k+1\mid k) + \Delta y = y_{k+1};$$

the estimate of the observation is the observation itself. With $c = 0$, (8.103) reduces to the Kalman filter.

We easily compute the recursion for the error covariance matrix associated with the estimator (8.103):

$$\begin{aligned} &P(k+1\mid k+1) \\ &\quad = P(k+1\mid k) - P(k+1\mid k)\,M^{T}(k+1) \\ &\qquad \times [M(k+1)\,P(k+1\mid k)\,M^{T}(k+1) + R(k+1)]^{-1} \\ &\qquad \times M(k+1)\,P(k+1\mid k) \\ &\qquad + \frac{c^{2}R^{2}(k+1)[M^{T}(k+1)\,M(k+1)]}{[M(k+1)\,P(k+1\mid k)\,M^{T}(k+1) + R(k+1)][M(k+1)\,M^{T}(k+1)]^{2}}. \end{aligned} \quad (8.104)$$

We note that the new term involving c^2 produces a desirable increase in P. Clearly, the appropriate value of c is not known *a priori*, and must be determined by simulations. In general, different values of c will be

required for each observation type. Schmidt, in the references cited, has successfully applied this filter in some interesting example problems.

The limited memory filter developed in Section7.10 falls in the present category of model error compensation techniques. Some simulations of the limited memory filter are presented in Section 12.

11. ADAPTIVE NOISE ESTIMATION

In Section 9, we described several systematic methods of determining the noise input levels (variances) as a means of compensating for model errors. All the methods described depend on simulation to establish appropriate values of certain parameters [e.g., k in (8.89), p in (8.91)]. In a simulation using real data, one looks at measurement residuals, such as (8.8), to evaluate filter performance. Residuals should be small, random, and should possess statistical properties consistent with (8.9) and (8.10). Clearly, we cannot be thinking here of real-time filtering. Such simulations can be performed only after the fact.

Instead of looking at residuals after the fact to evaluate filter performance, we propose to look at them in real time and to let the residuals themselves determine appropriate noise input levels. We let the noise input levels *adapt* to the residuals. Our development here is based on the work of Jazwinski [15, 17, 18].

Consider first the special case of *uncorrelated* and *identically distributed* noise inputs, that is, $Q(k) = qI$ in (8.1). We suppose that the noise coefficient matrix $\Gamma(k)$ has been engineered for us, perhaps by one of the methods described in Section 9. In this case, the noise variance term in the error covariance matrix prediction equation (8.3) is simply $q\Gamma(k)\,\Gamma^{\mathrm{T}}(k)$. We assume henceforth, and throughout this section, that measurements are scalar ($m = 1$). This is usually no loss of generality, since vector measurements can be filtered by processing their components one at a time.

Suppose that we are at time t_k, we have just processed observation y_k, and we are about to predict to t_{k+1} via (8.2) and (8.3). Let us use that value of the noise variance q which produces the *most probable* predicted residual $r(k+1 \mid k)$, as defined in (8.8), that is, determine the q value (call it $\hat{q}_{k,1}$) by the operation

$$\max_{q \geq 0} p[r(k+1 \mid k)], \tag{8.105}$$

where $p(\cdot)$ is the probability density function of the residual. The restriction $q \geq 0$ is consistent with the property of a variance.

Now the density in (8.105) is zero-mean, Gaussian, with the variance given in (8.10). It is easy to show [15] that the maximizing q is determined by

$$r^2(k+1 \mid k) = \mathscr{E}\{r^2(k+1 \mid k)\} \tag{8.106}$$

if

$$r^2(k+1 \mid k) > M(k+1)\,\Phi(k+1,k)\,P(k \mid k)\,\Phi^{\mathrm{T}}(k+1,k)\,M^{\mathrm{T}}(k+1) + R(k+1), \tag{8.107}$$

and is zero otherwise. Since

$$\mathscr{E}\{r^2(k+1 \mid k)\} = M(k+1)\,\Phi(k+1,k)\,P(k \mid k)\,\Phi^{\mathrm{T}}(k+1,k)\,M^{\mathrm{T}}(k+1) + qM(k+1)\,\Gamma(k)\,\Gamma^{\mathrm{T}}(k)\,M^{\mathrm{T}}(k+1) + R(k+1), \tag{8.108}$$

and if we use the slightly abused notation

$$\mathscr{E}\{r^2(k+1 \mid k) \mid q \equiv 0\} = M(k+1)\,\Phi(k+1,k)\,P(k \mid k) \times \Phi^{\mathrm{T}}(k+1,k)\,M^{\mathrm{T}}(k+1) + R(k+1), \tag{8.109}$$

our estimate of q is

$$\hat{q}_{k,1} = \begin{cases} \dfrac{r^2(k+1 \mid k) - \mathscr{E}\{r^2(k+1 \mid k) \mid q \equiv 0\}}{M(k+1)\,\Gamma(k)\,\Gamma^{\mathrm{T}}(k)\,M^{\mathrm{T}}(k+1)}, & \text{if positive} \\ 0 & \text{otherwise.}^{7} \end{cases} \tag{8.110}$$

The usual linear filter [(8.2)–(8.7)], with the estimator (8.110) used for q in (8.3), is our basic *adaptive filter* for uncorrelated and identically distributed noise inputs. Let us analyze the implications of the noise variance estimator (8.110). Except for an appropriate scaling factor, $\hat{q}_{k,1}$ is the excess, if any, of the residual squared over the expected value of the residual squared, under the assumption of no input noise. If the estimate $\hat{q}_{k,1}$ is positive, as determined by (8.110), then it produces the equality in (8.106). In any case, it produces *consistency* between residuals and their statistics.

Our adaptive filter is adaptive in the following sense. As long as residuals remain within their 1σ limits, the noise input level in the filter is zero. This is as it should be, since residuals are small and consistent with their statistics; the filter is operating satisfactorily. When residuals become large relative to their *predicted* 1σ values, the filter is diverging. $\hat{q}_{k,1}$ increases $P(k+1 \mid k)$, which increases the filter gain and "opens"

[7] We of course assume that $M\Gamma\Gamma^{\mathrm{T}}M^{\mathrm{T}} > 0$. That is, M is not orthogonal to Γ, which is an observability condition.

the filter to incoming observations. Subsequent filtering reduces $P(k+1 \mid k)$ again.

The adaptive filter we have described has a rather serious defect. The estimate of q is based on only one residual and therefore has little statistical significance. The estimator (8.110) will respond to (large) measurement noise samples as well as large residuals caused by model errors. This can be remedied in several ways. The simplest remedy is to smooth the estimates of q. Another remedy, which is described in detail by Jazwinski [17], is to replace the one predicted residual by the *sample mean* of N predicted residuals:

$$m_r \triangleq N^{-1} \sum_{l=1}^{N} r(k+l \mid k)/R^{1/2}(k+l).$$

The resulting estimator of q has the same structure as (8.110). This has been found very effective [17], but we shall not pursue the matter here. We note in passing that the present theory can also be applied to the estimation of the measurement noise variance R [15].

We next consider the more general case of *n independent* noise inputs in (8.1), that is, $\Gamma(k) = I$, Q $n \times n$ and diagonal. This structure obviates the requirement of engineering a noise coefficient matrix, which, as we noted in Section 9, is not a simple matter. Let us define the *predicted residuals*

$$r(k+l \mid k) = y_{k+l} - \mathscr{E}\{y_{k+l} \mid Y_k\}, \qquad l > 0. \tag{8.111}$$

Note that $r(k+1 \mid k)$ has already been defined in (8.8). These residuals are zero-mean, Gaussian, and it is straightforward to compute

$$\begin{aligned} r(k+l \mid k) &= M(k+l)\,\Phi(k+l, k)[x_k - \hat{x}(k \mid k)] \\ &\quad + M(k+l) \sum_{i=1}^{l} \Phi(k+l, k+i)\,\Gamma(k+i-1)\,w_{k+i} + v_{k+l} \end{aligned} \tag{8.112}$$

and

$$\begin{aligned} &\mathscr{E}\{r(k+l \mid k)\, r(k+m \mid k)\} \\ &\quad = M(k+l)\,\Phi(k+l, k)\,P(k \mid k)\,\Phi^{\mathrm{T}}(k+m, k)\,M^{\mathrm{T}}(k+m) \\ &\qquad + M(k+l) \sum_{i=1}^{l} [\Phi(k+l, k+i)\,\Gamma(k+i)\,Q\Gamma^{\mathrm{T}}(k+i) \\ &\qquad\qquad \times \Phi^{\mathrm{T}}(k+m, k+i)\,M^{\mathrm{T}}(k+m)] \\ &\qquad + R(k+l)\,\delta_{lm}, \qquad m \geq l. \end{aligned} \tag{8.113}$$

By analogy with the special case developed previously ($Q = qI$), and in particular, by analogy with (8.106), we determine a diagonal Q from the *consistency requirement*

$$r^2(k+l \mid k) = \mathscr{E}\{r^2(k+l \mid k)\}, \qquad l = 1,\ldots, N. \tag{8.114}$$

If (8.114) is satisfied, residuals will be consistent with their statistics. Equation (8.114) is related to the maximization of the joint density

$$p[r(k+1 \mid k),\ldots, r(k+N \mid k)],$$

but we do not pursue that relationship here. In our present case [$\Gamma(k) = I, Q$ diagonal], we have, from (8.113),

$$\mathscr{E}\{r^2(k+l \mid k)\} = \mathscr{E}\{r^2(k+l \mid k) \mid Q \equiv 0\} + \sum_{j=1}^{n} a_{lj} q_{jj}, \tag{8.115}$$

where

$$\mathscr{E}\{r^2(k+l \mid k) \mid Q \equiv 0\} = M(k+l)\,\Phi(k+l, k)\,P(k \mid k) \times \Phi^{\mathrm{T}}(k+l, k)\,M^{\mathrm{T}}(k+l) + R(k+l), \tag{8.116}$$

$$a_{lj} = \sum_{i=1}^{l} [M(k+l)\,\Phi(k+l, k+i)]_j^2, \tag{8.117}$$

and q_{jj} are the diagonal elements of Q. Introducing the vectors and matrix

$$\epsilon^{\mathrm{T}} = [r^2(k+1 \mid k) - \mathscr{E}\{r^2(k+1 \mid k) \mid Q \equiv 0\},\ldots, r^2(k+N \mid k) - \mathscr{E}\{r^2(k+N \mid k) \mid Q \equiv 0\}],$$

$$q^{\mathrm{T}} = [q_{11},\ldots, q_{nn}], \qquad A = [a_{lj}] \quad (N \times n),$$

the consistency requirement (8.114) becomes

$$Aq = \epsilon. \tag{8.118}$$

The solution of (8.118) can be written in general as

$$\bar{q}_{k,N} = (A^{\mathrm{T}}A)^{\#} A^{\mathrm{T}}\epsilon, \tag{8.119}$$

where $B^{\#}$ is the pseudo-inverse of B. Various special cases of (8.119), depending on N, are discussed by Jazwinski [15, 17]. Now, in case N is taken small, an estimate of q based on (8.119) may have little statistical significance. Various devices that give such an estimate statistical

significance are discussed by Jazwinski [17]. Here, motivated by least squares estimation theory, we make the *ad hoc* modification to (8.119):

$$\bar{q}_{k,N} = (C^{-1} + A^{\mathrm{T}}A)^{-1}(A^{\mathrm{T}}\epsilon + C^{-1}q^0). \tag{8.120}$$

q^0 is an *a priori* estimate of the vector q, and C^{-1} (diagonal) is a measure of belief in the *a priori* estimate. Based on (8.120), our estimator of q is

$$\hat{q}_{k,N} = \bar{q}_{k,N}, \tag{8.121}$$

subject to the rule that, if $(\hat{q}_{jj})_{k,N} < 0$, we set $(\hat{q}_{jj})_{k,N} = 0$. The extended Kalman filter (Theorem 8.1), with noise variance estimator (8.121), is the adaptive filter used in the simulations described below.

We present simulations of this adaptive filter in IMP[8]-type orbit estimation. The IMP satellite has a highly eccentric orbit that is extremely sensitive to model errors (see Wolf [41]). As a result, it represents an excellent test for the adaptive filter. The material presented below is taken from Jazwinski [18].

For the purpose of our adaptive filter simulation study, the following (true) Keplerian osculating orbit elements were assumed (at time zero):

$a = 16.37317$ earth radii (semimajor axis),

$e = 0.9370696$ (eccentricity),

$i = 33.33961$ deg (inclination),

$\omega = 133.1376$ deg (argument of perigee),

$\Omega = -101.6029$ deg (right ascension of ascending node),

$P = 93.29312$ hr (period),

$\nu = 1.492517$ rad (true anomaly),

(earth radius = 6378.165 km).

The dynamics used in the filter are two-body (Keplerian) dynamics, so that, as far as the filter dynamics are concerned, the orbital elements are constants of the motion. On the other hand, the true motion or satellite trajectory is generated via the theory given by Pines [25], which includes the secular effect of the second zonal harmonic in the earth's gravitational potential. The effect of this model error is a rigid rotation of the true orbit which causes a change in the argument of perigee (ω)

[8] Interplanetary Monitoring Probe, although IMP is an appropriate name for this satellite from the point of view of its orbit determination.

and the right ascension of the ascending node (Ω). In the case to be reported here, the stated model error produces $|\Delta\omega| = 0.0014$ and $|\Delta\Omega| = 0.0010$ at 747 hr. The other orbital elements are unaffected by this rotation. To summarize, the filter dynamics are two-body, whereas the real dynamics are not. Observations are generated on the basis of the real dynamics. The model error chosen is a realistic representation of the type of errors which are present in orbit estimation models. The initial estimation errors are

$$a - \hat{a} = -0.1336430; \qquad \omega - \hat{\omega} = -0.1266091;$$
$$e - \hat{e} = -0.0004206; \qquad \Omega - \hat{\Omega} = 0.0912469;$$
$$i - \hat{i} = 0.0004442; \qquad \nu - \hat{\nu} = 0.0013183;$$
$$P - \hat{P} = -1.144560;$$

and the initial estimation error covariance matrix, P_0, is consistent with these errors.

Observation types and times used in this study correspond to those of a real tracking schedule. Range, range rate, and l and m direction cosine observations were generated and contaminated with white Gaussian noise, with the respective variances: 0.676×10^{-7} earth radii, 0.529×10^{-7} earth radii/hr, 0.400×10^{-7}, 0.179×10^{-6}. The temporal distribution of observations is indicated in Fig. 8.11. A total of 1140

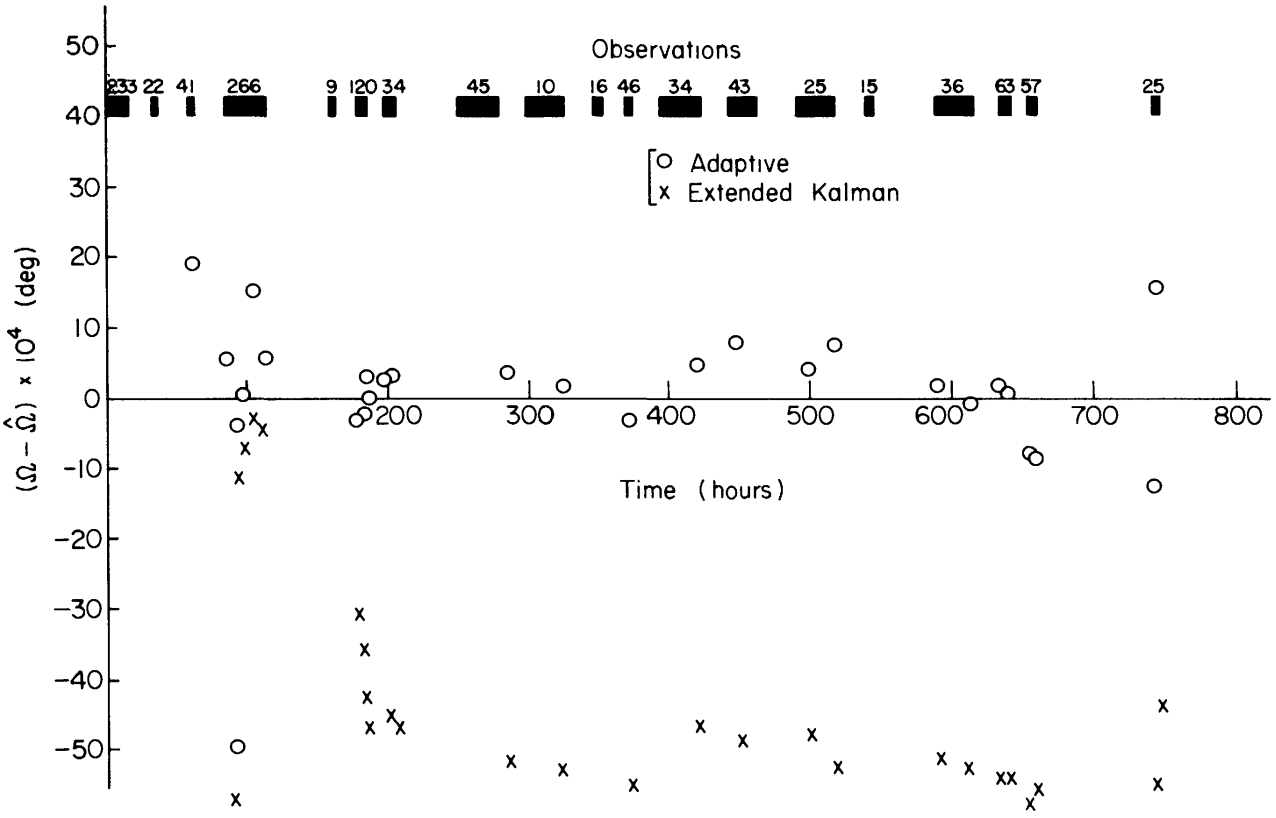


FIG. 8.11. Estimation errors in Ω.

observations were available, covering about eight revolutions of the satellite. Note that most of the observations were taken during the first two revolutions, and subsequent data are rather sparse.

We compare the performances of the extended Kalman filter and the adaptive filter. Both filters use two-body dynamics, but the adaptive filter estimates a (fictitious) noise covariance matrix, whereas the extended Kalman filter contains no system noise. We use six residuals in the adaptive filter ($N = 6$), $q^0 = 0$, and $C^{-1} = 10^{22}$ on the diagonal. The latter number is relatively easily arrived at after processing six observations. Fresh estimates of q are only computed every sixth observation.

Recall that the true orbit is rotated as described, producing a secular change in ω and Ω. The estimation errors[9] in ω and Ω, for the two filters, are shown in Figs. 8.11 and 8.12. Not shown are the initial transient

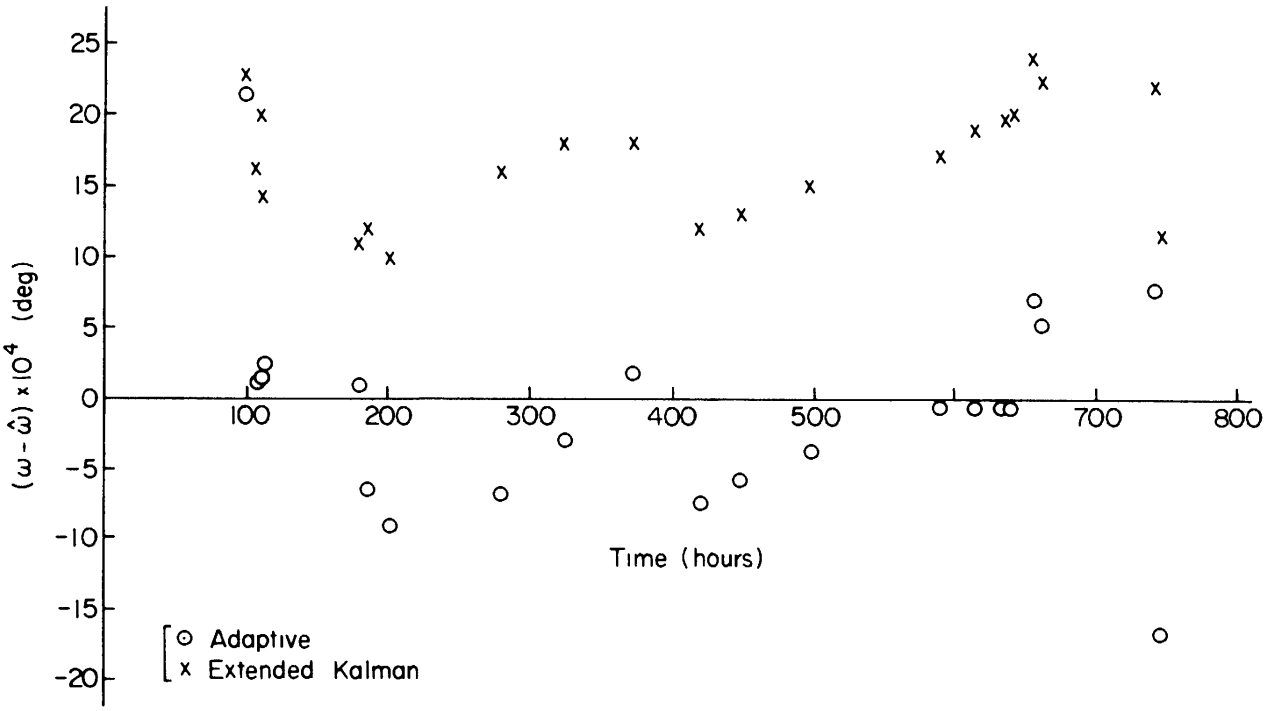


FIG. 8.12. Estimation errors in ω.

errors, similar for both filters, as they are off-scale. We see that, shortly after one revolution, the extended Kalman filter acquires the orbit quite well. Subsequently, however, the extended Kalman estimation errors drift off and are biased. The sense of the errors is precisely such that the extended Kalman filter is tracking the nonrotating (erroneous) orbit. The rms Kalman errors (square roots of the diagonal elements of the

[9] The errors plotted are representative; actual errors at each of the 1140 observations are not plotted.

P matrix) are small, the filter gain is small, and the extended Kalman filter is no longer correcting the orbit. In fact, the estimation errors range up to 10 standard deviations. The residuals are inconsistent with their predicted statistics. This is a typical example of filter divergence.

As is seen in Figs. 8.11 and 8.12, the adaptive filter acquires the orbit and tracks the (true) rotating orbit. Estimation errors are small and relatively random. The adaptive filter residuals are always consistent with their predicted statistics. That is, the estimation errors are generally less than one standard deviation, and almost always less than two standard deviations.

The approximate magnitudes of the errors in the other orbital elements at 747 hr are listed in the following tabulation.

	Extended Kalman	Adaptive
$\lvert a - \hat{a} \rvert$	1×10^{-7}	4×10^{-6}
$\lvert e - \hat{e} \rvert$	2×10^{-6}	5×10^{-7}
$\lvert i - \hat{i} \rvert$	1×10^{-3}	4×10^{-4}
$\lvert P - \hat{P} \rvert$	1×10^{-6}	4×10^{-5}
$\lvert \nu - \hat{\nu} \rvert$	2×10^{-5}	5×10^{-6}

We see that, in all the orbital elements but the period (or, equivalently, the semimajor axis), the adaptive filter errors are smaller than the extended Kalman filter errors. Now in the IMP orbit, the period is particularly sensitive to errors. Since many observations are taken in the first 1.3 revolutions, before the model errors cause the filter to diverge, the extended Kalman filter is able to obtain a very good estimate of the period. Subsequent model errors do not affect the period. On the other hand, because of the sensitivity, the uncertainties generated by the adaptive filter somewhat degrade the estimate of the period. This is probably legitimate, since uncertainties in the argument of perigee should degrade the knowledge of the period.

These and other simulations indicate that the adaptive filter is a promising model error compensation technique.

12. LIMITED MEMORY FILTERING

In this section, we present an application of the limited memory filter developed in Section 7.10 to orbit determination in the presence of model errors. We use the limited memory procedure described in the discussion of Theorem 7.10. That is, conditioning of the estimate on

old data is discarded in batches of N; the filter memory varies from N to $2N$. The data presented below are taken from Jazwinski [16].

Our dynamical system for the limited memory filtering study is the "rectilinear" orbit problem with dynamics

$$\ddot{x} = -(\mu/x^2) \qquad (x \text{ scalar}).$$

μ is the gravitational constant times earth mass, 19.9094165 earth radii^3/hr^2. In first-order form,

$$\dot{x}_1 = x_2\,, \qquad \dot{x}_2 = -(\mu/x_1^2),$$

where x_1 is position and x_2 velocity. Our observations consist of position (range) data

$$y_k = (x_1)_k + v_k\,,$$

with

$$\mathscr{E}\{v_k\} = 0, \qquad \mathscr{E}\{v_k v_l\} = r\delta_{kl}\,, \qquad r = 1.0 \times 10^{-7} \quad \text{earth radii}^2.$$

The model error is a bias in μ; observations are generated on the basis of the value of μ given in the foregoing, whereas the filter uses the value 19.9244165.

The true trajectory is a "rectilinear ellipse" starting at approximately 8 earth radii at zero time, going to apogee of 34 earth radii at 42 hr, and terminating at 26 earth radii at 70 hr. The range data were generated at the rate of 1 observation every 0.1 hour, but a gap in the data was left between 40 and 60 hours. The data gap will demonstrate how the limited memory filter performs when faced with prediction errors caused by system nonlinearity.

The performances of the extended Kalman filter and the limited memory filter will be compared on the basis of actual estimation errors and the RSS/RMS ratios. $RSS(k)$ is the square root of the time average of the squares of the estimation errors, whereas $RMS(k)$ is the square root of the appropriate diagonal element of the error covariance matrix. That is,

$$RSS(k) = \left\{k^{-1} \sum_{i=1}^{k} [x_l(i) - \hat{x}_l(i \mid i)]^2\right\}^{1/2},$$

$$RMS(k) = [P_{ll}(k \mid k)]^{1/2},$$

where $l = 1, 2$ for position and velocity, respectively. Clearly, $RSS(k)/RMS(k)$ should approach one for large k.

Figures 8.13 and 8.14 show how the extended Kalman filter behaves

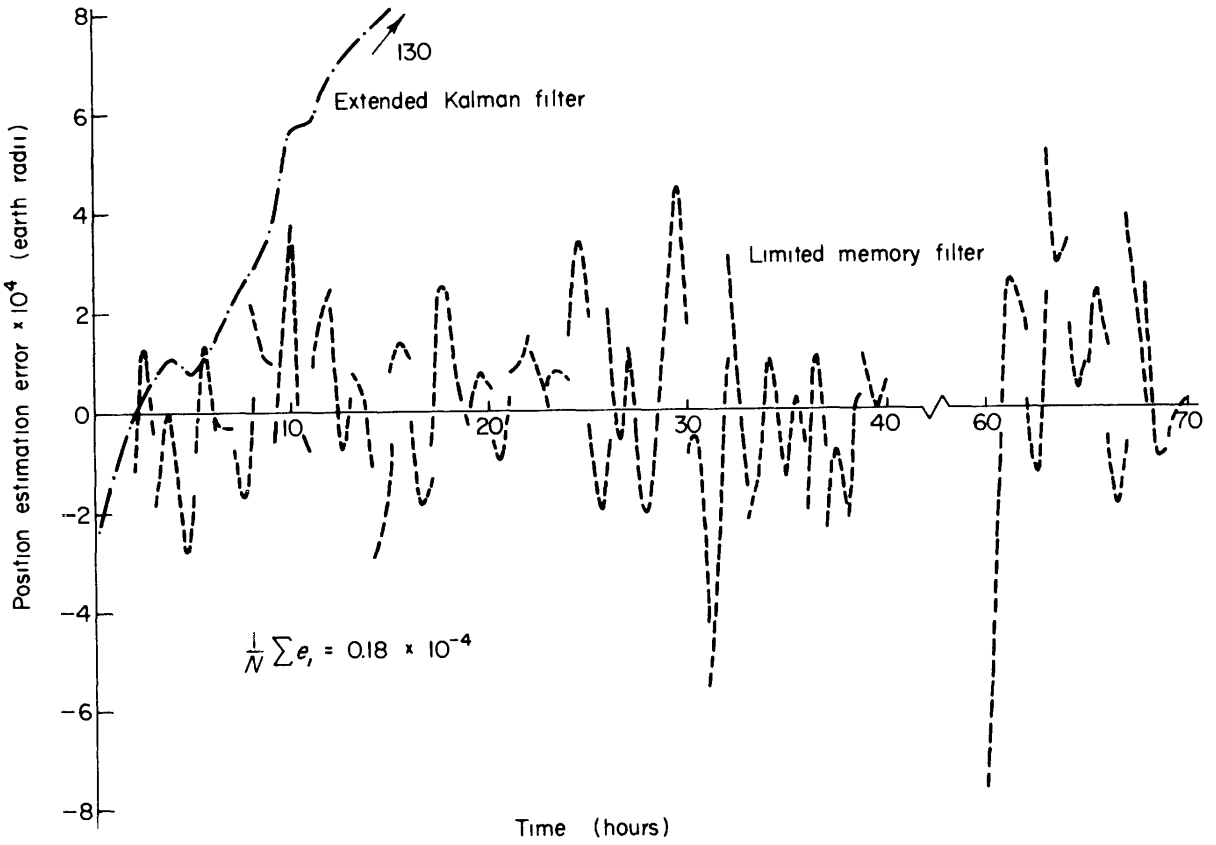


FIG. 8.13. Position estimation errors.

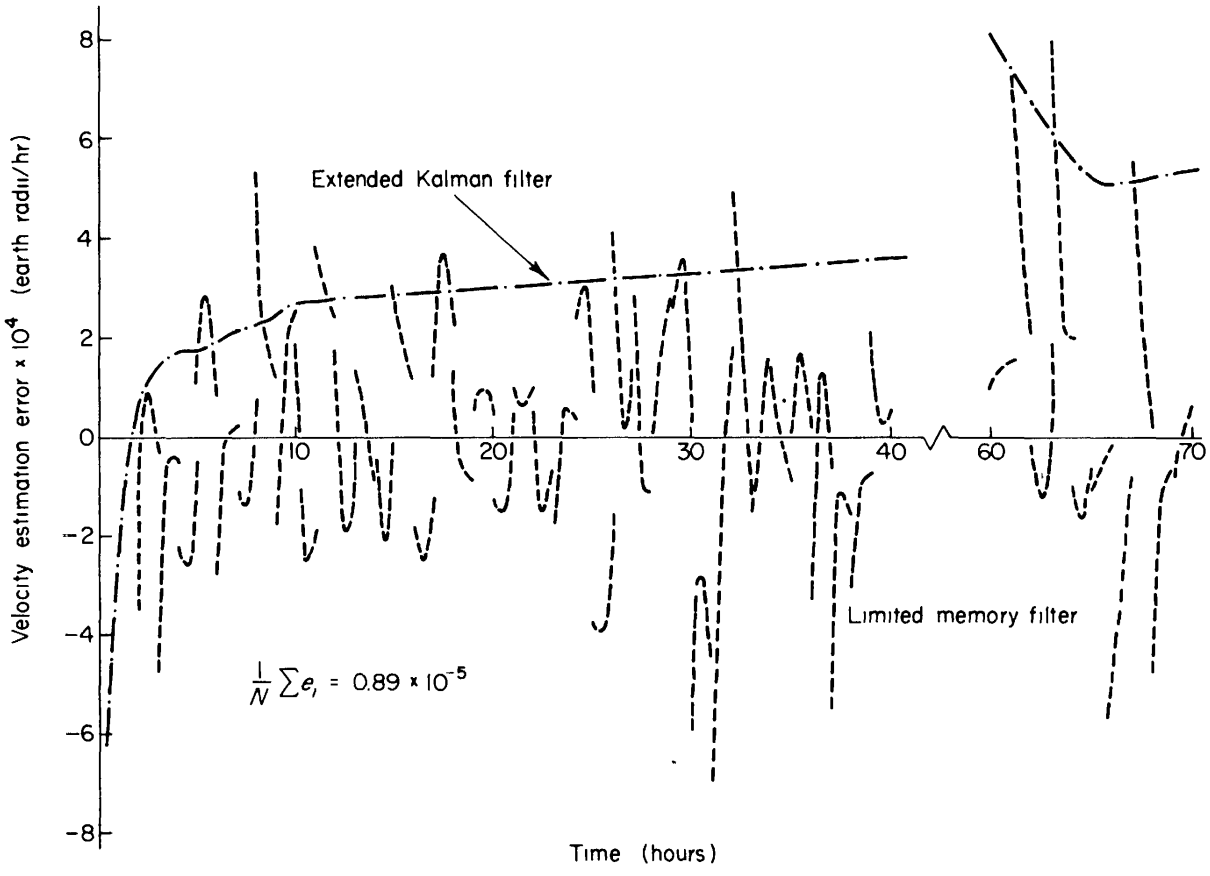


FIG. 8.14. Velocity estimation errors.

in terms of actual estimation error. The position estimation error, in particular, is seen to grow very rapidly. The velocity estimation error does not grow as much, but is definitely not random. Figures 8.15 and 8.16 show the histories of the *RSS/RMS* ratios for position and velocity. These are seen to grow almost exponentially in the extended Kalman

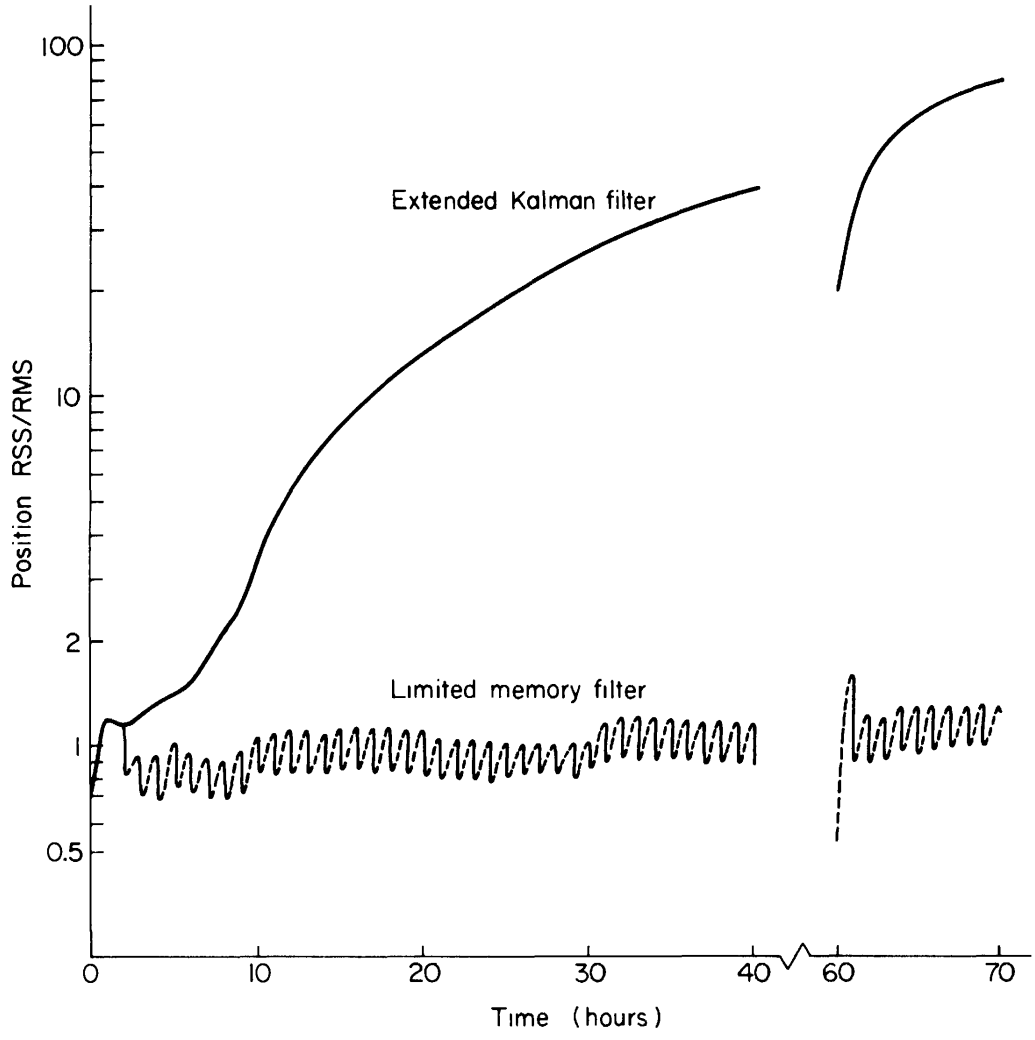


FIG. 8.15. Position *RSS/RMS* history

filter. Thus, in the presence of the bias, the Kalman error uncertainties quickly become small since many accurate data have been processed, and then the filter fails to track the orbit.

The limited memory filter performance is shown in the same figures. In these simulations, old data are discarded in batches of 10, and the filter memory varies between 10 and 20. Reference to Figs. 8.13 and 8.14 shows that the estimation errors are rather random, with appropriately small averages. The estimation errors remain within approximately 2 sigma measurement noise level, and the filter is tracking the orbit. More accurate determination of the orbit is not possible in view of the

Brownian motion process. Thus stochastic differential equations of the type we are studying are merely idealizations of physical processes, and can only serve to, in some sense, approximate a physical process. We recall (Section 3.8) that white noise approximates a physical process with short correlation times. The question arises: In what sense, if any, does the solution of a stochastic differential equation approximate a physical process?

To be specific, consider a physical process that is modeled by the scalar differential equation

$$\dot{y}_t = f(y_t, t) + g(y_t, t)m_t, \tag{4.78}$$

where m_t is a random Gaussian disturbance. Let $\{m_t\}$ be a zero-mean, exponentially correlated [see (3.107)] stationary process with correlation function

$$\gamma^{\rho}(t + \tau, t) = \mathcal{E}\{m_{t+\tau}m_t\} = \sigma^2(\rho/2)\, e^{-\rho|\tau|}. \tag{4.79}$$

We saw (Section 3.8) that, for ρ large, $\{m_t\}$ is a wide-band process that approximates white noise. We note that if $\{m_t\}$ is correlated, then $\{y_t\}$ is not a Markov process. The $\{y_t\}$ process is, however, smooth. The correlation function (4.79) is integrable, so that $\{y_t\}$ is differentiable; Eq. (4.78) is a well-defined differential equation for the sample functions.

Now the correlated process $\{m_t\}$, sometimes called *colored* noise, can be generated by a linear stochastic differential equation forced by white noise (see Example 4.12),

$$dm_t = -\rho m_t\, dt + \sigma\rho\, d\beta_t, \tag{4.80}$$

where $\{\beta_t\}$ is a Brownian motion process with unit variance parameter, and ρ, $\sigma > 0$ are fixed constants. Taking (4.78) and (4.80) together,

$$\begin{aligned} dy_t &= [f(y_t, t) + g(y_t, t)m_t]\, dt \equiv h(y_t, m_t, t)\, dt, \\ dm_t &= -\rho m_t\, dt + \sigma\rho\, d\beta_t, \end{aligned} \tag{4.81}$$

we have a vector stochastic differential equation, with state vector

$$x_t^{\mathrm{T}} = [y_t, m_t],$$

of the type studied in Section 4. $\{x_t\}$ is a Markov process. Thus (4.78) can be modeled by a stochastic differential equation.[7] We have, however,

[7] In this section, we reserve the term *stochastic differential equation* to mean the equation of Section 4, be it interpreted in the Itô or Stratonovich sense. Equations such as (4.78) are models for physical processes.

increased the dimension of the problem. Instead of a scalar equation (4.78), we now have a two-dimensional system (4.81).

Example 4.12. Consider the linear stochastic differential equation

$$dm_t = -\rho m_t\, dt + \sigma\rho\, d\beta_t\,, \qquad t \geqslant 0, \tag{4.82}$$

where $\{\beta_t\,, t \geqslant 0\}$ is a Brownian motion process with unit variance parameter, and ρ, $\sigma > 0$ are fixed constants. Let $m_0 \sim N(0, \sigma^2\rho/2)$, independent of $\{\beta_t\}$. We saw in Section 6 that (4.82) is the same in the Itô and Stratonovich sense (since the coefficient of $d\beta_t$ is nonrandom), and can be manipulated by formal rules as if β_t were continuously differentiable. We therefore consider the equivalent equation

$$\dot{m}_t = -\rho m_t + \sigma\rho w_t\,, \qquad t \geqslant 0, \tag{4.83}$$

where $\{w_t\}$ is zero-mean, white Gaussian noise with

$$\mathscr{E}\{w_t w_\tau\} = \delta(t - \tau).$$

The solution of (4.83) is

$$m_t = m_0 e^{-\rho t} + \sigma\rho \int_0^t e^{-\rho(t-s)} w_s\, ds. \tag{4.84}$$

Clearly,

$$\mathscr{E}\{m_t\} \equiv 0.$$

Compute $(t > \tau)$

$$\begin{aligned}
\mathscr{E}\{m_t m_\tau\} &= \mathscr{E}\left\{\left[m_0 e^{-\rho t} + \sigma\rho \int_0^t e^{-\rho(t-s)} w_s\, ds\right]\right. \\
&\qquad \left. \times \left[m_0 e^{-\rho\tau} + \sigma\rho \int_0^\tau e^{-\rho(\tau-\xi)} w_\xi\, d\xi\right]\right\} \\
&= \sigma^2(\rho/2) e^{-\rho(t+\tau)} + \sigma^2\rho^2 \int_0^t \int_0^\tau e^{-\rho(t-s)} e^{-\rho(\tau-\xi)}\, \delta(s - \xi)\, ds\, d\xi \\
&= \sigma^2(\rho/2) e^{-\rho(t+\tau)} + \sigma^2\rho^2 \int_0^\tau e^{-\rho(t-\xi)} e^{-\rho(\tau-\xi)}\, d\xi \\
&= \sigma^2(\rho/2) e^{-\rho(t-\tau)}.
\end{aligned} \tag{4.85}$$

Thus (4.82) generates a stationary, exponentially correlated, Gaussian process. $\{m_t\}$ is sometimes called *colored* noise. ▲

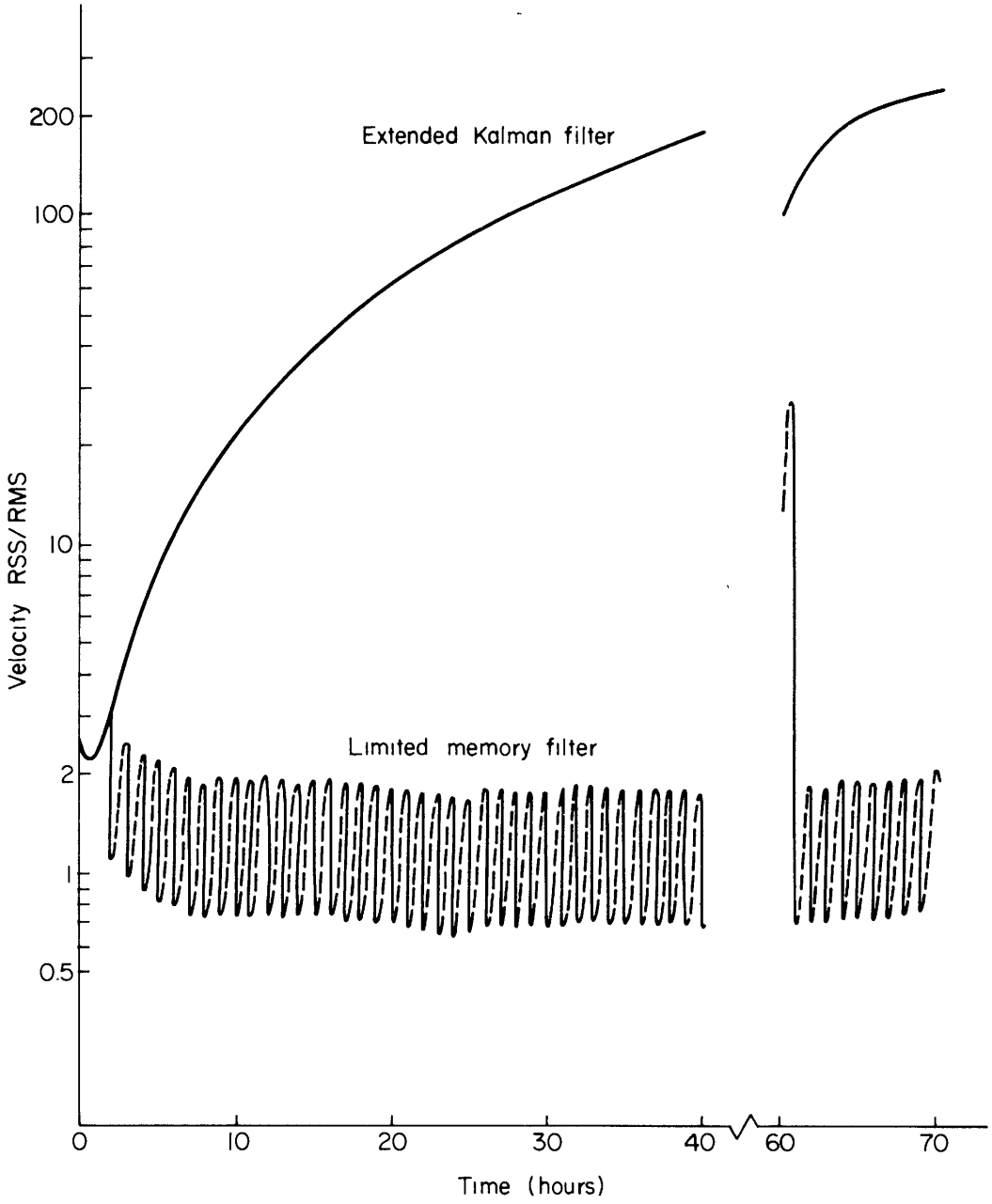


FIG. 8.16. Velocity *RSS/RMS* history.

μ bias. More accurate determination would require the estimation of the bias itself.

Figures 8.15 and 8.16 contain several interesting features. First, it is seen that the limited memory filter covariance matrix is consistent with the estimation errors ($RSS/RMS < 3$), except at 60 hours in Fig. 8.16, which will be discussed later. Note that the *RSS/RMS* ratios always grow and then decrease the instant old data are discarded. Apparently the filter begins to diverge; the model is not quite accurate enough over time-arcs of 2 hr. Note the unusual *RSS/RMS* ratios at 60 hr. Apparently, the computation of the covariance matrix over the data gap was inaccurate because of system nonlinearity. Thus at 60 hours the

covariance matrix is in error. This error is eliminated, together with old data, at 61 hours. The erroneous covariance matrix is discarded in the same way as is an initial condition (see discussion following Theorem 7.10). The limited memory filter is clearly computationally stable.

These simulations indicate that the limited memory filter is effective in model error compensation. No extensive study of this filter has yet been made, however. In any given estimation problem, it is necessary to determine the best memory length of the filter. The memory length must be, of course, related to the effective errors in the model, as well as the amount of data per unit time which is available. The idea of limited memory filtering actually goes to the heart of the real problem in estimation. For a given required estimation accuracy, our dynamical model must be sufficiently accurate, say over $t_1 \leqslant t \leqslant t_2$, so that the data in that interval can give us the required estimation accuracy. Conversely, given the best model available, adequate over a given time interval, there must be enough data available in that interval to produce the desired estimation accuracy.

13. MISCELLANEOUS TOPICS

There are a number of interesting topics in linear filtering theory which we do not cover in this book. These topics are peripheral to the basic theory but are, nevertheless, important in certain applications. We briefly outline some of these topics here.

Throughout this book, we suppose that a measurement schedule is prespecified for us, and all that remains is to filter the data. In designing a system, however, part of the design is to engineer a measurement subsystem, and to decide what kind of data and how many data will be made available to the filter. Here we run into various tradeoffs, since a measurement system costs money. Various optimization problems thus arise. One such problem is the following. Given a fixed total number of observation times, what is the optimum observation schedule that minimizes *rms* estimation errors? Conversely, given a desired *rms* estimation error, what is the minimum number and schedule of observations? These and related problems have been treated by Denham and Speyer [8] and by Potter and Fagan [26], for example. These authors give additional references on the subject.

A review of the linear filter computational procedure (Section 2) quickly reveals that the necessary filter computations rapidly increase with the dimension (n) of the dynamical system. The larger n is, the larger the computer required for the filter. This has motivated some

research on the design of suboptimal filters of dimension less than n. In this category is the work of Johansen on the minimal-order observer, and the partitioning approach of Joseph. This and related work are reviewed by Aoki [2].

The computation of the estimation error covariance matrix P is often plagued with computational error (round-off) problems. These problems are particularly acute when much information is available. For then the covariance matrix may decrease by many orders of magnitude. The typical numerical problem of subtracting small and almost equal numbers arises. This problem can clearly be solved by using higher precision arithmetic. An interesting alternative is to compute the *square root* of P rather than P itself. Recursions for $P^{1/2}$ have been developed by Bellantoni and Dodge [6], and also by Andrews [1]. In essence, one can maintain the same precision of $P^{1/2}$ in single-precision arithmetic as one can maintain using P in double-precision arithmetic. We also note the related work of Potter and Fraser [27] which develops recursions for the determinant of P, which is useful in some error analysis applications.

APPENDIX 8A

ORBIT MECHANICS

The purpose of this appendix is to summarize briefly the mechanics of motion of an orbiting (earth) satellite. An extensive treatment of this subject may be found in [3].

The motion of a satellite of negligible mass in a point-mass gravitational field is described by the second-order differential equations

$$\ddot{R} = -(\mu/r^3)R, \tag{8A.1}$$

where

$$R^{\mathrm{T}} = [x\ y\ z], \qquad r = \| R \| = [R^{\mathrm{T}}R]^{1/2}$$

x, y, and z are the position coordinates of the satellite in an inertial coordinate system centered at the point-mass (center of the earth). μ is the (earth's) gravitational constant. Thus $\dot{R}$ is the velocity vector, and $\ddot{R}$ is the acceleration vector ($v \triangleq \| \dot{R} \|$).

Equations (8A.1) describe the so-called *two-body* motion, and we sometimes refer to the dynamics (8A.1) as two-body dynamics. As is well known, the satellite orbit is an ellipse, as shown in Fig. 8A.1. The point F is the focus of the ellipse containing the point-mass; C is the center of the ellipse or concentric circle of radius a. The ellipse (orbit)

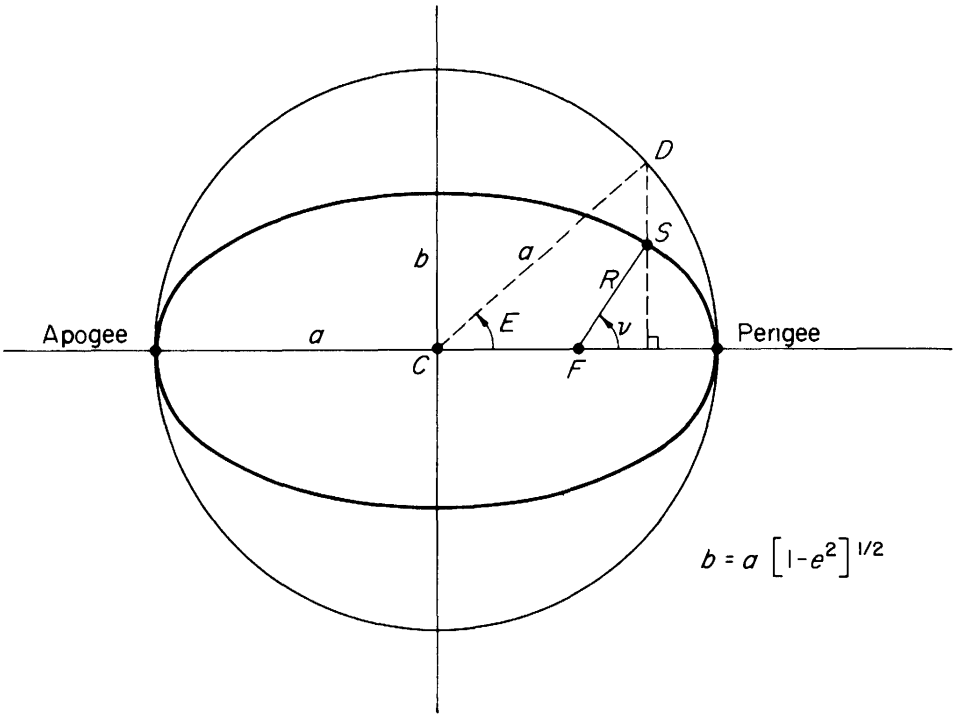


FIG. 8A.1. Satellite orbit plane.

is defined by its *semimajor axis a* and *eccentricity e*. The position of the satellite S in the orbit can be defined in terms of one of three angles. The *true anomaly* v is the angle subtended at the focus by the line of apsides (perigee-apogee) and the radius vector R. The *eccentric anomaly* E is the angle subtended at the center of the ellipse by the line of apsides and the line CD. The *mean anomaly* M is defined by *Kepler's equation*

$$M = E - e \sin E. \tag{8A.2}$$

These angles are measured positive from perigee in the direction of the satellite's motion. The (perigee-to-perigee) *period* p and the *mean motion* n of the satellite are functions of the semimajor axis (and μ) given by

$$p = 2\pi a^{3/2}\mu^{-1/2}, \tag{8A.3}$$

$$n = \mu^{1/2}a^{-3/2}. \tag{8A.4}$$

It remains to describe the orientation of the satellite orbit in inertial space. This is done via three angles as shown in Fig. 8A.2. First, the inertial coordinate system, centered at the earth's center (point-mass), is defined as follows. The x axis passes through the vernal equinox, and the (x, y) plane is the earth's equatorial plane. The z axis passes through the earth's north pole. The y axis completes a right-handed coordinate system. The intersection of the orbit plane with earth's equatorial plane is called the line of nodes (FN). Then, the *inclination* i

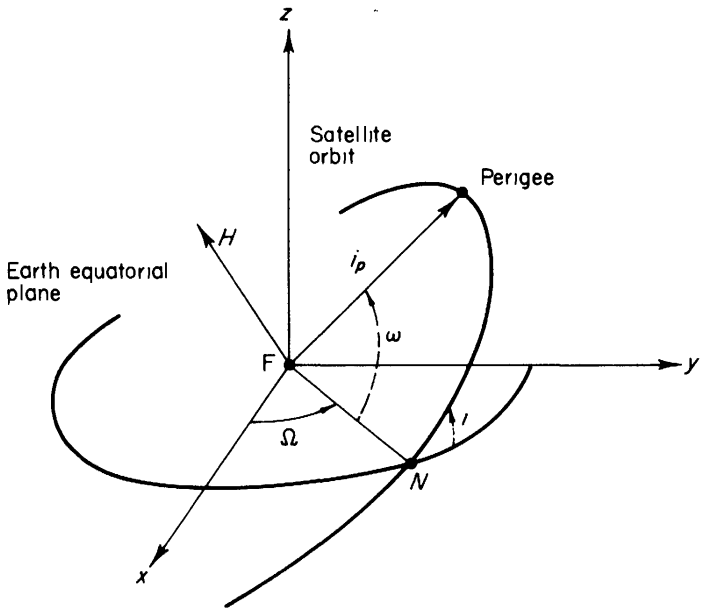


FIG. 8A.2. Inertial orientation of satellite orbit plane.

is the angle between the orbit plane and earth's equatorial plane. The *argument of perigee* ω, measured in the orbit plane, is the angle between the line of nodes and the line of apsides. Finally, *the right ascension of the ascending node* Ω, measured in the plane of the equator, is the angle between the x-axis and the line of nodes (positive east to the ascending node).

To summarize, the six (independent) orbital elements of the satellite's motion are

$$\begin{array}{ll} a \text{ (or } p, \text{ or } n), & i, \\ e, & \omega, \\ \nu \text{ (or } E, \text{ or } M), & \Omega. \end{array} \tag{8A.5}$$

They are called the *Keplerian orbital elements* and, at any time t, uniquely define the position (R) and velocity ($\dot{R}$) of the satellite. If we define the angular momentum vector

$$H = R \times \dot{R} \quad \text{(vector cross-product)}, \qquad h = \| H \|, \tag{8A.6}$$

with components (H_x, H_y, H_z), and the vector i_p from the focus to perigee (see Fig. 8A.2),

$$ei_p = \left\{\frac{1}{r} - \frac{1}{a}\right\} R - \frac{d}{\mu}\dot{R}, \qquad (d = R^T\dot{R}), \tag{8A.7}$$

then the orbital elements are given in terms of position and velocity by

$$
\begin{aligned}
a &= \left\{\frac{2}{r} - \frac{v^2}{\mu}\right\}^{-1}, \\
e\cos E &= 1 - r/a \qquad [M = E - e\sin E], \\
e\sin E &= d/(\mu a)^{1/2} \qquad [r = a(1-e^2)/(1+e\cos\gamma)], \\
\cos i &= H_z/h \quad \text{(first, fourth quadrant)}, \\
\sin\Omega &= H_x/h\sin i, \qquad \cos\Omega = -H_y/h\sin i, \\
\cos\omega &= (i_p)_x\cos\Omega + (i_p)_y\sin\Omega, \qquad \sin\omega = (i_p)_z/\sin i.
\end{aligned}
\tag{8A.8}
$$

Now the earth is not a point-mass and, as a result, (8A.1) is an incomplete description of earth satellite motion. Accelerations a_j, caused by the earth's gravitational harmonics, act on the satellite, and therefore

$$\sum a_j$$

must be added to the right-hand side of (8A.1). These accelerations cause both periodic and secular variations in the elements of the satellite's orbit. Earth oblateness, for example, causes a secular variation in the argument of perigee (ω) and in the right ascension of the ascending node (Ω), in addition to producing periodic variations in the other elements. The secular variations in ω and Ω, produced by the earth's second zonal harmonic, are used in [25] to simulate dynamical model errors in orbit determination. We use this device to demonstrate the performance of the adaptive filter in Section 11.

Additional innumerable accelerations act on the earth satellite. These include the gravitational accelerations of the sun, moon, and planets, solar radiation pressure, and drag caused by the earth's atmosphere. It is no wonder that the dynamical modeling problem is an extremely difficult one in earth satellite orbit determination, and that filter divergence is a real problem in this field (see Section 8).

We next summarize the radar observations most commonly used in orbit determination. These are *range* ρ,

$$\rho = [(x - x_s)^2 + (y - y_s)^2 + (z - z_s)^2]^{1/2}, \tag{8A.9}$$

where (x, y, z) is the satellite position and (x_s, y_s, z_s) is the position of the radar station; *range rate* $\dot{\rho}$,

$$\dot{\rho} = \rho^{-1}[(x - x_s)(\dot{x} + \Omega_E y_s) + (y - y_s)(\dot{y} - \Omega_E x_s) + (z - z_s)\dot{z}], \tag{8A.10}$$

where Ω_E is the earth's rotation rate; and the l and m *direction cosines.* The m direction cosine is the cosine of the angle subtended at the station between the horizon looking north and the satellite as it pierces the meridian plane containing the station. The l direction cosine is the cosine of the angle subtended at the station between the horizon looking east and the satellite as it pierces the plane of the parallel of latitude containing the station. There are many errors involved in the mathematical models for observations. These include errors in station location, atmospheric refraction that affects the radar signals, and so on. These, of course, compound the modeling problem mentioned earlier.

APPENDIX 8B

REENTRY MECHANICS

The coordinate system commonly used in reentry trajectory analysis is shown in Fig. 8B.1. The reentry vehicle state in three-dimensional reentry can be taken to be

$$x^{\mathrm{T}} = [v, \gamma, \psi, h, \varphi, \theta],$$

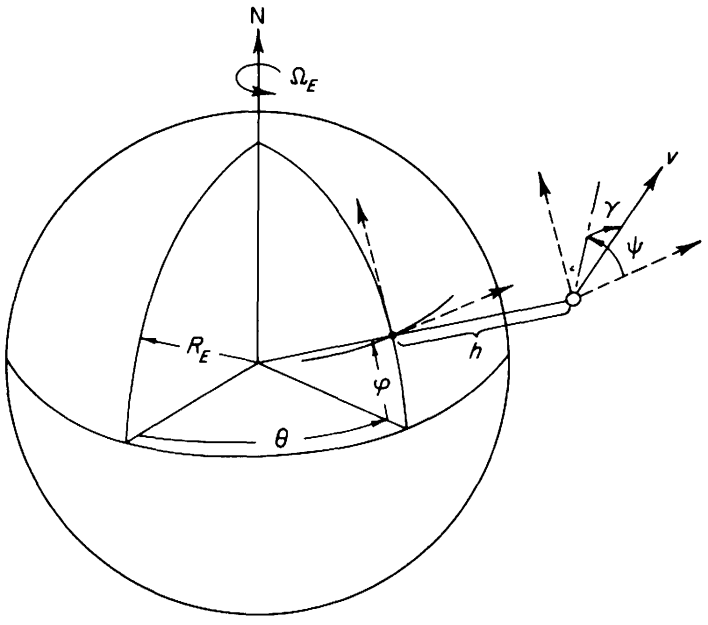


FIG. 8B.1. Reentry coordinate system.

where v is velocity (magnitude), γ is the flight path angle measured from the local horizontal, ψ is the heading angle measured from east toward north in the horizon plane, h is the altitude ($r \triangleq h + R_E$), φ the latitude, and θ the longitude.

In this coordinate system, the motion of the vehicle is described by the differential equations

$$\begin{aligned}
\dot{v} &= \frac{F_{xv}}{mv} - \frac{\mu}{r^2}\sin\gamma + \Delta\dot{v}_{\text{rot}}, \\
\dot{\gamma} &= \frac{F_{zv}}{mv} + \frac{v\cos\gamma}{r} - \frac{\mu}{vr^2}\cos\gamma + \Delta\dot{\gamma}_{\text{rot}}, \\
\dot{\psi} &= \frac{F_{yv}}{mv} - \frac{v}{r}\cos\gamma\cos\psi\tan\varphi + \Delta\dot{\psi}_{\text{rot}}, \\
\dot{h} &= v\sin\gamma, \\
\dot{\varphi} &= \frac{v}{r}\cos\gamma\sin\psi, \\
\dot{\theta} &= \frac{v}{r\cos\varphi}\cos\gamma\cos\psi.
\end{aligned} \tag{8B.1}$$

In these equations, μ is the earth's gravitational constant and m the vehicle mass (constant). F_{xv}, F_{yv}, and F_{zv} are the aerodynamic forces; and $\Delta\dot{v}_{\text{rot}}$, $\Delta\dot{\gamma}_{\text{rot}}$, and $\Delta\dot{\psi}_{\text{rot}}$ are due to earth's rotation. Earth oblateness has been neglected in (8B.1). Thus our reentry model corresponds to a *three-dimensional, constant-mass* reentry about a *spherical, rotating* earth.

The physical and mathematical details of this reentry model may be found in [39]. We mention here that the aerodynamic forces depend, among other things, on the orientation of the reentry vehicle relative to the wind. In case of a maneuverable lifting reentry vehicle, this orientation can be controlled. In the numerical experiments of Section 7, these controls are specified.

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